

MATH 3220-002 Selected Solutions to PSet Problems

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In this document you may find solutions (proofs or examples) to selected problems from problem sets. As a general rule of thumb, it is more likely that this document will be updated with the solutions to problems that were graded on accuracy.

Throughout this class your solutions need not match the level of polish, precision and detail of the below solutions. What's most important is grasping the underlying concepts and steadily improving your problem-solving skills as well as your proof-writing and presentation skills.

Most solutions will have follow-up exercises. Just as in the problem sets, these follow-up exercises are stated in a neutral language, and determining whether or not the statement is correct is part of the exercise. A follow-up exercise here may later show up in an exam.

1 PSet 1, Problem 3

Let $f : V \rightarrow W$ be an arbitrary function between vector spaces.
Then f is linear iff its graph is a linear subspace of $V \times W$.

This statement is true without any modifications:

Claim: Let $f : V \rightarrow W$ be an arbitrary function between vector spaces. Then f is linear iff its graph is a linear subspace of $V \times W$.

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Proof: First note that the **Cartesian product** $V \times W$ of the vector spaces $V = (V, +_V, \cdot_V)$ and $W = (W, +_W, \cdot_W)$ itself is a vector space, with addition

$$+_{V \times W} : (V \times W) \times (V \times W) \rightarrow V \times W$$

and scalar multiplication

$$\cdot_{V \times W} : \mathbb{R} \times (V \times W) \rightarrow V \times W$$

defined entrywise, that is, for any $(v_1, w_1), (v_2, w_2) \in V \times W$, the vector sum is by definition

$$(v_1, w_1) +_{V \times W} (v_2, w_2) = (v_1 +_V v_2, w_1 +_W w_2),$$

and similarly, for any $(v, w) \in V \times W$ and for any scalar $c \in \mathbb{R}$, scalar multiplication is by definition

$$c \cdot_{V \times W} (v, w) = (c \cdot_V v, c \cdot_W w).$$

Thoroughout we'll use this linear structure on $V \times W$. As is customary, for brevity we'll suppress subscripts for vector addition and we'll not use any specific symbols for scalar multiplication.

($\Rightarrow$) First suppose $f : V \rightarrow W$ is linear. We claim that its graph is

a vector subspace of $V \times W$, that is, $\text{graph}(f)$ is closed under vector addition and scalar multiplication.

Let $(v_1, w_1), (v_2, w_2) \in \text{graph}(f)$, $c \in \mathbb{R}$. It suffices¹ to show that

$$c(v_1, w_1) + (v_2, w_2) = (cv_1 + v_2, cw_1 + w_2) \in \text{graph}(f).$$

By the definition of the graph of a function, this is true iff

$$cw_1 + w_2 = f(cv_1 + v_2).$$

Furthermore, we had taken the vectors $(v_1, w_1), (v_2, w_2)$ to be in the set $\text{graph}(f)$, so that we also have

$$w_1 = f(v_1), \quad w_2 = f(v_2).$$

Combining these equations with the previous one we obtain

$$cw_1 + w_2 = cf(v_1) + f(v_2) = f(cv_1 + v_2),$$

where in the last equality we used the linearity of f .

($\Leftarrow$) Now suppose $f : V \rightarrow W$ is an arbitrary function such that $\text{graph}(f)$ is a vector subspace of $V \times W$. We claim that f is in fact linear. For this it suffices² to show that for any $v_1, v_2 \in V$ and for any $c \in \mathbb{R}$, we have

$$f(cv_1 + v_2) = cf(v_1) + f(v_2).$$

By the definition of the graph of a function, we know that

$$(v_1, f(v_1)), (v_2, f(v_2)) \in \text{graph}(f),$$

and since we are assuming $\text{graph}(f)$ to be a linear subspace, it

¹See Follow-up Exercise 1 below.

²By Follow-up Exercise 1 below.

must be closed under linear combinations, hence

$$(cv_1 + v_2, cf(v_1) + f(v_2)) = c(v_1, f(v_1)) + (v_2, f(v_2)) \in \text{graph}(f).$$

Again by the definition of the graph of a function, it must be the case that

$$f(cv_1 + v_2) = cf(v_1) + f(v_2).$$

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This statement leads to a more general idea: what type of properties of a function can one recover by investigating its graph instead, and under which conditions? For instance the statement we established guarantees that the linearity of a function is equivalent to the linearity of its graph. Such statements in general are called "closed graph theorems"³.

³See Terence Tao's [blog post](#) surveying closed graph theorems in different contexts.

Follow-up Exercise 1

- (i) Let V and W be vector spaces, $f : V \rightarrow W$ be a function. Then f is linear iff

$$\forall v_1, v_2 \in V, \forall c \in \mathbb{R} : f(cv_1 + v_2) = cf(v_1) + f(v_2).$$

- (ii) Let V be a vector space, $S \subseteq V$ be a subset. Then S is a linear subspace iff

$$\forall v_1, v_2 \in S, \forall c \in \mathbb{R} : cv_1 + v_2 \in S.$$

Follow-up Exercise 2 Let V and W be vector spaces. Then

- (i) There are nonlinear functions $f : V \rightarrow W$ such that $\text{graph}(f)$ is closed under vector addition but not necessarily scalar multiplication.
- (ii) There are nonlinear functions $f : V \rightarrow W$ such that $\text{graph}(f)$ is closed under scalar multiplication but not necessarily vector addition.

Follow-Up Exercise 3 Let V and W be vector spaces. Then

- (i) Any proper linear subspace X of $V \times W$ is the graph of some function $f : V \rightarrow W$. Here **proper** means $X \neq V \times W$.
- (ii) Any proper linear subspace of V is the graph of some function $f : L \rightarrow R$ for some linear subspaces L, R of V such that **for any vector $v \in V$, there are unique $l \in L$ and $r \in R$ with $v = l + r$.**

2 PSet 2, Problem 3

Let V, W and X be normed spaces. Then

- (i) For any $L \in \mathcal{L}(V; W)$ and for any $K \in \mathcal{L}(W; X)$, $|K \circ L| \leq |K||L|$.
- (ii) If V, W , and X are finite dimensional, then $1 = \inf\{C \in \mathbb{R}_{\geq 0} \mid \forall L, K : |K \circ L| \leq C|K||L|\}$.

Let us examine the two parts separately. We'll use subscripts for different norms to disambiguate different magnitudes.

The first part is correct without any modifications:

Claim: Let V, W and X be normed spaces. Then for any $L \in \mathcal{L}(V; W)$ and for any $K \in \mathcal{L}(W; X)$, $|K \circ L|_{\text{op}} \leq |K|_{\text{op}}|L|_{\text{op}}$.

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Proof: Fix two linear transformations $L : V \rightarrow W$ and $K : W \rightarrow X$. Note that by problem 2.a in **PSET1**, $K \circ L : V \rightarrow X$ too is a linear operator. If either K or L is the zero transformation, then the LHS in the inequality is zero, hence there is nothing to prove. If neither K nor L is zero, and further either K or L is an unbounded linear operator, the RHS in the inequality is ∞ , and there is nothing to prove again.

Thus suppose both K and L are bounded linear operators different from the zero transformation. By a discussion in class, we know that for any $v \in V$ and for any $w \in W$,

$$|L(v)|_W \leq |L|_{\text{op}}|v|_V, \quad |K(w)|_X \leq |K|_{\text{op}}|w|_W.$$

Let $v \in V$. Then

$$|(K \circ L)(v)|_X = |K(L(v))|_X \leq |K|_{\text{op}}|L(v)|_W \leq |K|_{\text{op}}|L|_{\text{op}}|v|_V.$$

Thus by the definition of the operator norm, we have

$$|K \circ L|_{\text{op}} \leq |K|_{\text{op}} |L|_{\text{op}}.$$

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Next we discuss the second part. Let us first consider the Euclidean case, that is, the normed spaces involved are all spaces of tuples of real numbers, with norm induced by the standard dot product. In this case, say considering the standard bases, we can simply take the linear operators to be matrices of appropriate dimension. We'll also assume that all spaces involved are of at least dimension 1; indeed if any one of the spaces involved is zero dimensional (thus consisting of the zero vector only), then the statement fails (see below).

Claim: Let $V = \mathbb{R}^n$, $W = \mathbb{R}^m$ and $X = \mathbb{R}^p$ for n, m, p positive integers. Then

$$\inf \left\{ C \in \mathbb{R}_{\geq 0} \mid \forall L \in \mathcal{M}(m \times n; \mathbb{R}), \forall K \in \mathcal{M}(p \times m; \mathbb{R}): \begin{array}{l} |KL|_{\text{op}} \leq C |K|_{\text{op}} |L|_{\text{op}} \end{array} \right\} = 1,$$

where the operator norms are induced by norms defined by dot products.

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Proof: Denote by α the infimum in the claim. Thus the claim is that $\alpha = 1$. The first claim already established that $\alpha \leq 1$. Thus it suffices to show $\alpha \geq 1$.

Suppose $\alpha < 1$. We'll obtain a contradiction by constructing two transformations L and K such that

$$|K \circ L|_{\text{op}} = 1, \quad |K|_{\text{op}} = 1, \quad |L|_{\text{op}} = 1.$$

Define L to be the $m \times n$ matrix such that, if L_{ij} is the ij -th entry of

L , then $L_{ij} = 1$ if $i = 1 = j$ and $L_{ij} = 0$ otherwise. Similarly define K to be the $p \times m$ matrix with 1 in the top left entry and zero otherwise. Then a straightforward calculation shows that KL is the $p \times n$ matrix with 1 in the top left entry and zero in all other entries.

Let us now compute the operator norms of K , L and KL . Note that in fact **wlog** it suffices to compute the operator norm of one of these matrices. Let $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$. Then

$$|Lv| = |\underbrace{(v_1, 0, 0, \dots, 0)}_{\in \mathbb{R}^m}| = |v_1| \leq \left(\sum_{i=1}^n |v_i|^2 \right)^{1/2} = |v|.$$

Hence $|L|_{\text{op}} \leq 1$. Further, for $v = (1, 0, \dots, 0) \in \mathbb{R}^n$, we have $|Lv| = 1 = |v|$, so that $|L|_{\text{op}} = 1$. Similarly the operator norms of both matrices K and KL are 1. By the definition of α , we have

$$1 = |KL|_{\text{op}} \leq \alpha |K|_{\text{op}} |L|_{\text{op}} = \alpha \leq 1,$$

a contradiction. ┘

A word regarding the construction used in the previous proof is in order. The construction is based on the heuristic that the operator norm of a linear transformation is the maximal stretch it can cause. In the above proof, we considered L and K to be the transformations that "delete" all but the first entries. This means that, effectively these two transformations "stretch" vectors along the "first axes" by a factor of 1, and completely collapse all other directions.

Let us next consider the "trivial" case:

Claim: Let $V = \mathbb{R}^n$, $W = \mathbb{R}^m$ and $X = \mathbb{R}^p$ for n, m, p nonnegative integers. If any one of n, m , or p is zero, then

$$\inf \left\{ C \in \mathbb{R}_{\geq 0} \mid \forall L \in \mathcal{M}(m \times n; \mathbb{R}), \forall K \in \mathcal{M}(p \times m; \mathbb{R}): \begin{array}{l} |KL|_{\text{op}} \leq C |K|_{\text{op}} |L|_{\text{op}} \end{array} \right\} = 0,$$

where the operator norms are induced by norms defined by dot products.

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Proof: We have three cases to consider: $n = 0$, $m = 0$, or $p = 0$. In any one of these cases, $K \circ L$ is necessarily the zero transformation, and since the operator norm is a norm, $\|K \circ L\|_{\text{op}} = 0$ and hence the infimum is zero.

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Finally we discuss the general version of the second part of the statement. While this too is true (assuming that none normed spaces are zero dimensional), with what we have developed so far we can not prove it as is. Instead we'll prove a version of the second part of the statement with two additional hypotheses (which hypothesis we'll later see in the course is redundant). The idea is similar to the construction we used before, namely, we'll use two rank 1 transformations whose stretch effects compound maximally. Indeed, ultimately the two additional hypotheses are there so that the construction for matrices work in the abstract case:

Claim: Let V, W , and X be finite but positive dimensional normed spaces. If

- there are nonzero bounded linear functions $V \rightarrow \mathbb{R}$ and $W \rightarrow \mathbb{R}$, and
- W has the property that

$$\forall \kappa \in \mathcal{BL}(W; \mathbb{R}), \exists w_0 \in W : \|w_0\|_W = 1, |\kappa(w_0)| = \|\kappa\|_{\text{op}},$$

then

$$\inf \left\{ C \in \mathbb{R}_{\geq 0} \mid \forall L \in \mathcal{L}(V; W), \forall K \in \mathcal{L}(W; X) : \frac{\|KL\|_{\text{op}}}{\|K\|_{\text{op}}\|L\|_{\text{op}}} \leq C \right\} = 1.$$

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Proof: We already have by the first claim that 1 is an upper bound on the infimum involved. Let $x_0 \in X$ be a nonzero vector, $\kappa : W \rightarrow \mathbb{R}$ be nonzero bounded linear (note that we assume there is such a κ) and define $K : W \rightarrow X$ by

$$K : w \mapsto \kappa(w)x_0.$$

Then due to the linear nature of κ , K is linear. Further, a straightforward calculation shows that

$$|K|_{\text{op}} = |\kappa|_{\text{op}}|x_0|_X.$$

By hypothesis, there is a $w_0 \in W$ such that

$$|w_0|_W = 1, \quad |\kappa(w_0)| = |\kappa|_{\text{op}}.$$

Let $\lambda : V \rightarrow \mathbb{R}$ be nonzero bounded linear (again we assume such a λ exists) and define $L : V \rightarrow W$ by

$$L : v \mapsto \lambda(v)w_0.$$

L too is linear and again by a calculation we have

$$|L|_{\text{op}} = |\lambda|_{\text{op}}|w_0|_W = |\lambda|_{\text{op}}.$$

Consequently we have

$$|K|_{\text{op}}|L|_{\text{op}} = |\kappa|_{\text{op}}|\lambda|_{\text{op}}|x_0|_X.$$

On the other hand, for $v \in V$ arbitrary,

$$K \circ L(v) = K(\lambda(v)w_0) = \lambda(v)K(w_0) = \lambda(v)\kappa(w_0)x_0 = \lambda(v)|\kappa|_{\text{op}}x_0.$$

Consequently,

$$|K \circ L(v)|_X = |\lambda(v)| |\kappa|_{\text{op}} |x_0|_X.$$

Optimizing over all $v \in V$, we obtain

$$|K \circ L|_{\text{op}} = |\lambda|_{\text{op}} |\kappa|_{\text{op}} |x_0|_X = |K|_{\text{op}} |L|_{\text{op}},$$

establishing that the infimum involved indeed is equal to 1.

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In closing, let us mention why the two additional hypotheses are redundant: First, any linear operator defined on a finite dimensional normed space turns out to be a bounded linear operator. Second, for any linear operator defined on a finite dimensional normed space, it turns out that there is an input vector with unit norm whose image under the linear operator has norm exactly the operator norm of the operator.

Follow-Up Exercise 1 Let n be a positive integer and $\mathcal{M}$ be the vector space of $n \times n$ matrices with real entries. Then for any norm $|\bullet|$ on $\mathcal{M}$, the following holds:

$$\forall A, B \in \mathcal{M} : |AB| \leq |A||B|.$$

Follow-Up Exercise 2 Let n be a positive integer and $\mathcal{M}$ be the vector space of $n \times n$ matrices with real entries. Then for any norm $|\bullet|$ on $\mathcal{M}$, the following holds:

$$\exists c \in \mathbb{R}_{\geq 0}, \forall A, B \in \mathcal{M} : |AB| \leq c|A||B|.$$

Follow-Up Exercise 3 Let n be a positive integer and $\mathcal{M}$ be the vector space of $n \times n$ matrices with real entries. Then

$$\exists A, B \in \mathcal{M} : |AB|_{\text{op}} \leq |A|_{\text{op}}|B|_{\text{op}}.$$

Follow-Up Exercise 4 Let V, W and X be normed spaces. Then

$$1 = \inf\{C \in \mathbb{R}_{\geq 0} \mid \forall L, K : |K \circ L| \leq C|K||L|\}.$$

3 PSet 3, Problem 2

Let V be a normed space and $v_\bullet$ be a sequence in V converging to 0. Then for any normed space W and any bounded sequence of bounded linear operators $B_\bullet$ from V to W , $\lim_{n \rightarrow \infty} B_n(v_n) = 0$.

This statement is indeed true without any modifications:

Claim: Let V be a normed space and $v_\bullet$ be a sequence in V converging to 0. Then for any normed space W and any bounded sequence of bounded linear operators $B_\bullet$ from V to W , $\lim_{n \rightarrow \infty} B_n(v_n) = 0$.

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Proof: For any n , we have by a result we proved in class,

$$0 \leq |B_n(v_n)|_W \leq |B_n|_{\text{op}} |v_n|_V \leq \underbrace{\left(\sup_n |B_n|_{\text{op}} \right)}_{< \infty} |v_n|_V.$$

Note that we have that $\sup_n |B_n|_{\text{op}}$ is a finite number by the assumption that $B_\bullet$ is a *bounded sequence* of bounded operators. As $v_\bullet$ converges to the zero vector in V , by the continuity of norms, the sequence $|v_\bullet|_V$ of scalars converges to the zero number. Thus by the fact that a scalar multiple of a sequence of scalars that converges to the zero number converges to zero itself, and the squeeze theorem, we are done.

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Follow-Up Exercise Let V and W be normed spaces and $B_\bullet$ be a sequence of bounded linear operators from V to W converging to 0. Then for any bounded sequence $v_\bullet$ in V , $\lim_{n \rightarrow \infty} B_n(v_n) = 0$.

4 PSet 4, Problem 2

Let X be a metric space and $A, B \subseteq X$. Then

- (i) $\text{clos}(A \cup B) = \text{clos}(A) \cup \text{clos}(B)$.
- (ii) $\text{clos}(A \cap B) = \text{clos}(A) \cap \text{clos}(B)$.
- (iii) $\text{int}(A \cup B) = \text{int}(A) \cup \text{int}(B)$.
- (iv) $\text{int}(A \cap B) = \text{int}(A) \cap \text{int}(B)$.

The set operations of closure and interior have many different (but equivalent) definitions. Throughout the discussion below let us use the definitions given in class⁴. Thus if $A \subseteq X$ is a subset of a metric space, then the closure $\text{clos}(A)$ of A is by definition the set of all those points in X that can be approximated arbitrarily well by points in A , and the interior $\text{int}(A)$ of A is by definition the set of all those points in A whose sufficiently small perturbations are in A . Formally, we have:

$$\begin{aligned}\text{clos}(A) &= \{x \in X \mid \forall \varepsilon \in \mathbb{R}_{>0} : X[x] < \varepsilon \cap A \neq \emptyset\}, \\ \text{int}(A) &= \{x \in X \mid \exists \varepsilon \in \mathbb{R}_{>0} : X[x] < \varepsilon \subseteq A\}.\end{aligned}$$

Recall that here $X[x] < \varepsilon$ is the open ball in X centered at x or radius ε , that is,

$$X[x] < \varepsilon = [x] < \varepsilon = \{x' \in X \mid d(x', x) < \varepsilon\}.$$

With these definitions in mind, let's investigate the four statements given. The first statement is correct without any modification:

⁴While many parts of the first problem in this problem set too can be used to study the statements in this problem, we decided to present a self-contained account to exemplify the type of argumentation based on first principles.

Claim: Let X be a metric space. Then

$$\forall A, B \subseteq X : \text{clos}(A \cup B) = \text{clos}(A) \cup \text{clos}(B).$$

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Proof: Fix two subsets $A, B \subseteq X$. To show the equality of LHS and RHS it suffices to show that LHS is a subset of RHS and RHS is a subset of LHS.

($\supseteq$) Let $x \in \text{clos}(A) \cup \text{clos}(B)$. Then either x is in the closure of A , or x is in the closure of B (or both). Thus either for any $\varepsilon \in \mathbb{R}_{>0}$, $X[x] < \varepsilon \cap A$ is nonempty, or for any $\varepsilon \in \mathbb{R}_{>0}$, $X[x] < \varepsilon \cap B$ is nonempty. Then in either case for any $\varepsilon \in \mathbb{R}_{>0}$,

$$\emptyset \subsetneq (X[x] < \varepsilon \cap A) \cup (X[x] < \varepsilon \cap B) = X[x] < \varepsilon \cap (A \cup B),$$

as intersections distribute over unions. Thus x is in the closure of $A \cup B$.

($\subseteq$) We'll argue **contrapositively**. Let $x \notin \text{clos}(A) \cup \text{clos}(B)$. Then by **de Morgan's laws**, x is neither in the closure of A nor is it in the closure of B . Thus there are $\alpha, \beta \in \mathbb{R}_{>0}$ such that

$$X[x] < \alpha \cap A = \emptyset = X[x] < \beta \cap B.$$

Put $\varepsilon = \min\{\alpha, \beta\}$. Then again by the distributivity of intersections over unions

$$\begin{aligned} X[x] < \varepsilon \cap (A \cup B) &= (X[x] < \varepsilon \cap A) \cup (X[x] < \varepsilon \cap B) \\ &= \emptyset \cup \emptyset = \emptyset, \end{aligned}$$

so that x is not in the closure of $A \cup B$.

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The second statement is not quite correct; indeed in general it works only in one direction:

Claim: Let X be a metric space. Then

$$\forall A, B \subseteq X : \text{clos}(A \cap B) \subseteq \text{clos}(A) \cap \text{clos}(B).$$

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Proof: Fix two subsets $A, B \subseteq X$. Let x be in the closure of $A \cap B$. Thus by the monotonicity of intersections, for any $\varepsilon \in \mathbb{R}_{>0}$,

$$\emptyset \subsetneq X[x] < \varepsilon \cap (A \cap B) \subseteq X[x] < \varepsilon \cap A,$$

so that x is in the closure of A , as well as

$$\emptyset \subsetneq X[x] < \varepsilon \cap (A \cap B) \subseteq X[x] < \varepsilon \cap B,$$

so that x is in the closure of B also.

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Claim: There are subsets A, B in some metric space X such that

$$\text{clos}(A \cap B) \subsetneq \text{clos}(A) \cap \text{clos}(B).$$

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Proof: To prove this claim it suffices to exhibit an example. For instance consider $X = \mathbb{R}$, $A = [-1, 0[$, $B =]0, 1]$. Then $A \cap B$ is empty, hence so is its closure. On the other hand the closure of A is $[-1, 0]$ and the closure of B is $[0, 1]$, so that the intersection of the closures of A and B is precisely the **singleton** $\{0\}$.

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The next two statements provided are about the interior operation,

and we can study them either using the definitions or **duality**:

Lemma 1 (Duality): Let X be a metric space. Then

$$\forall A \subseteq X : \text{int}(A^c) = \text{clos}(A)^c.$$

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Here, for $B \subseteq X$, $B^c = X \setminus B$ is the **complement** of the set B inside X ; thus in **set builder notation**, $B^c = \{x \in X \mid x \notin B\}$. Note that the duality lemma extends the duality between unions and intersections established by complementation, that is, the de Morgan laws, to the interior and closure operations.

Proof (of Duality Lemma): Fix a subset $A \subseteq X$. By careful argumentation we can show the equality of two sets by chaining logical equivalences as follows:

$$\begin{aligned} x \in \text{int}(A^c) & \\ \Leftrightarrow \exists \varepsilon \in \mathbb{R}_{>0} : X[x] < \varepsilon \subseteq A^c & \\ \Leftrightarrow \exists \varepsilon \in \mathbb{R}_{>0} : X[x] < \varepsilon \cap A = \emptyset & \\ \Leftrightarrow x \notin \text{clos}(A) & \\ \Leftrightarrow x \in \text{clos}(A)^c. & \end{aligned}$$

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Claim: Let X be a metric space. Then

$$\forall A, B \subseteq X : \text{int}(A \cap B) = \text{int}(A) \cap \text{int}(B).$$

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We'll prove this claim in two ways: first using the duality lemma,

and second using the definition of interior.

Proof (using duality): Let $A, B \subseteq X$. Then we have

$$\begin{aligned}
 \text{int}(A \cap B) &\stackrel{*}{=} \text{int} \left((A \cap B)^c \right)^c \\
 &\stackrel{\dagger}{=} \text{clos} \left((A \cap B)^c \right)^c \\
 &\stackrel{\diamond}{=} \text{clos} \left(A^c \cup B^c \right)^c \\
 &= \left(\text{clos} \left(A^c \right) \cup \text{clos} \left(B^c \right) \right)^c \\
 &\stackrel{\diamond}{=} \text{clos} \left(A^c \right)^c \cap \text{clos} \left(B^c \right)^c \\
 &\stackrel{\dagger}{=} \text{int} \left(A^{cc} \right) \cap \text{int} \left(B^{cc} \right) \\
 &\stackrel{*}{=} \text{int}(A) \cap \text{int}(B),
 \end{aligned}$$

where in the equalities marked with $\star, \dagger, \diamond$ we used the double complement producing the original set, duality lemma, and de Morgan's law, respectively, and in the unmarked equality we used the distributivity of closure over unions we established before.

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Proof (using definitions): Fix two subsets $A, B \subseteq X$.

($\subseteq$) Let x be in the interior of $A \cap B$. Then for some $\varepsilon \in \mathbb{R}_{>0}$, $X[x] < \varepsilon$ is a subset of $A \cap B$. Since $A \cap B$ is a subset of A , by the **transitivity** of the "subset of" relation, $X[x] < \varepsilon$ is a subset of A , so that x is in the interior of A , and similarly this same open ball is a subset of B , so that x is in the interior of B also. Consequently $x \in \text{int}(A) \cap \text{int}(B)$.

($\supseteq$) Let $x \in \text{int}(A) \cap \text{int}(B)$. Thus x is in the interior of both A and B . Thus there are $\alpha, \beta \in \mathbb{R}_{>0}$ such that

$$X[x] < \alpha] \subseteq A, \quad \text{and} \quad X[x] < \beta] \subseteq B.$$

Put $\varepsilon = \min\{\alpha, \beta\}$. Then

$$X[x] < \varepsilon] \subseteq X[x] < \alpha] \cap X[x] < \beta] \subseteq A \cap B,$$

so that x is in the interior of $A \cap B$.

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The remaining statement is not quite correct:

Claim: Let X be a metric space. Then

$$\forall A, B \subseteq X : \text{int}(A \cup B) \supseteq \text{int}(A) \cup \text{int}(B).$$

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While we could prove this claim using duality again, let's argue using the definitions only:

Proof: Fix $A, B \subseteq X$. Let $x \in \text{int}(A) \cup \text{int}(B)$. Then x is either in the interior of A , or it is in the interior of B (or both). In the first case, for some $\alpha \in \mathbb{R}_{>0}$, $X[x] < \alpha] \subseteq A \subseteq A \cup B$, and similarly in the second case for some $\beta \in \mathbb{R}_{>0}$, $X[x] < \beta] \subseteq B \subseteq A \cup B$, so that in either case we are able to find an open ball centered at x completely contained in $A \cup B$, establishing that x is in the interior of $A \cup B$.

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Claim: There are subsets A, B in some metric space X such that

$$\text{int}(A \cup B) \subsetneq \text{int}(A) \cup \text{int}(B).$$

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Proof: It suffices to exhibit an example. By duality, the complements of the subsets in our previous example ought to work. Indeed, say

$X = \mathbb{R}$, $A^c = [-1, 0[$, $B^c =]0, 1]$, so that

$$A =]-\infty, -1[\cup [0, \infty[; \quad B =]-\infty, 0] \cup]1, \infty[.$$

Then $A \cup B = \mathbb{R}$, hence so is its interior. On the other hand the interior of A is $]-\infty, -1[\cup]0, \infty[$ and the interior of B is $]-\infty, 0[\cup]1, \infty[$, so that the union of the interiors of A and B is precisely $\mathbb{R} \setminus \{0\}$.

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Follow-up Exercise 1 There are metric spaces X such that

$$\forall A, B \subseteq X : \text{clos}(A \cap B) = \text{clos}(A) \cap \text{clos}(B).$$

Follow-up Exercise 2 Let X be a metric space. Then for any subset $S \subseteq X$, there are subsets $A, B \subseteq X$ such that

$$S = (\text{clos}(A) \cap \text{clos}(B)) \setminus \text{clos}(A \cap B)$$

Follow-up Exercise 3 Let A be a set and for any $\alpha \in A$, let S_α be a subset of a metric space X . Then

- (i) $\text{clos}(\bigcup_{\alpha \in A} S_\alpha) = \bigcup_{\alpha \in A} \text{clos}(S_\alpha).$
- (ii) $\text{clos}(\bigcap_{\alpha \in A} S_\alpha) = \bigcap_{\alpha \in A} \text{clos}(S_\alpha).$
- (iii) $\text{int}(\bigcup_{\alpha \in A} S_\alpha) = \bigcup_{\alpha \in A} \text{int}(S_\alpha).$
- (iv) $\text{int}(\bigcap_{\alpha \in A} S_\alpha) = \bigcap_{\alpha \in A} \text{int}(S_\alpha).$

Follow-Up Exercise 4 Let X be a metric space. Then

- (i) If two open balls in X intersect, then their intersection contains an open ball.
- (ii) For any $x, y \in X$, any $\alpha, \beta \in \mathbb{R}_{>0}$ and any $z \in [x] < \alpha] \cap [y] < \beta]$:

$$\begin{aligned} \sup\{\gamma \in \mathbb{R}_{>0} \mid [z] < \gamma] \subseteq [x] < \alpha] \cap [y] < \beta]\} \\ = \min\{\alpha - d(z, x), \beta - d(z, y)\}. \end{aligned}$$

- (iii) For any $x, y \in X$ and any $\alpha, \beta \in \mathbb{R}_{>0}$, consider the function

$$\gamma = \gamma_{x,y,\alpha,\beta} : X \rightarrow \mathbb{R}_{\geq 0}, z \mapsto \min\{\alpha - d(z, x), \beta - d(z, y)\}.$$

Then $[x] < \alpha$ and $[y] < \beta$ has a nonempty intersection iff $\gamma_{x,y,\alpha,\beta}$ is not identically zero.

5 PSet 5, Problem 3

Let $M = \{(x, y) \in \mathbb{R}^2 \mid |x| \leq 10, |y| \leq 10\}$ and consider

$f : M \rightarrow \mathbb{R}^3$ defined by

$$(x, y) \mapsto (xy - 2x + y + 1, x^2 + y^2 + x - 3y + 6, xy^2).$$

Perform the following tasks:

- (i) Compute the derivative of f .
- (ii) Compute the derivative of f at $x = (1, -1)$.
- (iii) Compute the affine function $A : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ tangent to f at $(1, -1)$.
- (iv) Compute the directional derivative of f at $x = (1, -1)$ in the direction of $v = (15, 2)$.
- (v) Compute the derivative of f' .
- (vi) Compute the derivative of $(f')'$.

Let us denote the components of f by f_1, f_2 and f_3 , so that

$$f_1 : (x, y) \mapsto xy - 2x + y + 1,$$

$$f_2 : (x, y) \mapsto x^2 + y^2 + x - 3y + 6,$$

$$f_3 : (x, y) \mapsto xy^2.$$

M is open Recall that for differentiability we require (at least for now) that the function at hand is defined on an open subset of a normed space. First let us verify that indeed M is an open subset of $\mathbb{R}^2$. Let $(x, y) \in M$, so that $-10 \leq x \leq 10$ and $-10 \leq y \leq 10$. We need to show that for some $\delta \in \mathbb{R}_{>0}$, the open δ -ball centered at (x, y) is in M . Put

$$\delta = \min\{10 - |x|, 10 - |y|\} = 1 - \max\{|x|, |y|\} > 0,$$

and suppose $(\bar{x}, \bar{y}) \in \mathbb{R}^2$ is in the open δ -ball centered at (x, y) . Then

$$\begin{aligned} |\bar{x}| - |x| &\leq ||\bar{x}| - |x|| \\ &\leq |\bar{x} - x| \\ &= \sqrt{(\bar{x} - x)^2} \\ &\leq \sqrt{(\bar{x} - x)^2 + (\bar{y} - y)^2} \\ &\leq \min\{10 - |x|, 10 - |y|\} \\ &\leq 10 - |x|, \end{aligned}$$

where in the second inequality we used the reverse triangular inequality. Cancelling $|x|$ from both sides we obtain $|\bar{x}| \leq 10$. Similarly we obtain $|\bar{y}| \leq 10$ also, so that $(\bar{x}, \bar{y}) \in M$.

Partial Derivatives Let us first record all partial derivatives of all components as we are familiar with from computation-based calculus classes. With higher order partial derivatives, it is likely that the order of differentiation will not matter, and we'll systematize the computation of partials like so: For each component f_j , we'll start a new chart. We'll put f_j to the bottom left corner. Then right next to it we'll put $\frac{\partial f_j}{\partial x}$ and right above it we'll put $\frac{\partial f_j}{\partial y}$. This way we'll populate the chart, one step to the right representing taking the x -partial and one step up representing taking the y -partial. Thinking of $\mathbb{Z}_{\geq 0}^2$ as a subset of the plane $\mathbb{R}^2$, this means that at the point $\alpha = (\alpha_1, \alpha_2)$ we have the $(\alpha_1 + \alpha_2)$ -th order partial derivative $\frac{\partial^{\alpha_1 + \alpha_2} f_j}{\partial x^{\alpha_1} \partial y^{\alpha_2}}$.

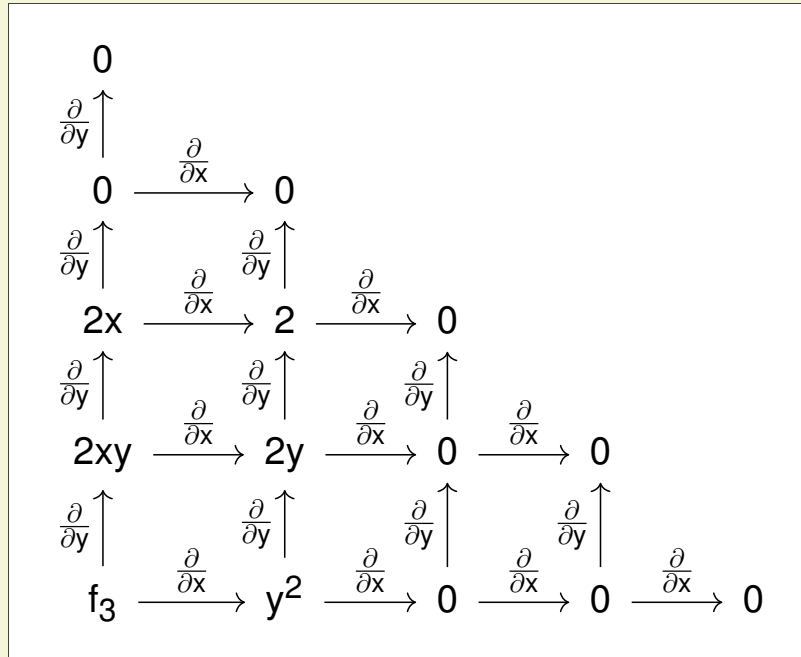
Thus the chart of partials of f_1 is:

$$\begin{array}{ccccccc}
 & & 0 & & & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial x} \rightarrow & & & \\
 & 0 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial x} \rightarrow & \\
 x+1 & \xrightarrow{\frac{\partial}{\partial x}} & 1 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & \\
 f_1 & \xrightarrow{\frac{\partial}{\partial x}} & y-2 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & \xrightarrow{\frac{\partial}{\partial x}} & 0
 \end{array}$$

Similarly the chart of partials of f_2 is:

$$\begin{array}{ccccccc}
 & & 0 & & & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial x} \rightarrow & & & \\
 & 2 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial x} \rightarrow & \\
 2y-3 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & \xrightarrow{\frac{\partial}{\partial x}} & 0 & & \\
 & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & & \frac{\partial}{\partial y} \uparrow & \\
 f_2 & \xrightarrow{\frac{\partial}{\partial x}} & 2x+1 & \xrightarrow{\frac{\partial}{\partial x}} & 2 & \xrightarrow{\frac{\partial}{\partial x}} & 0
 \end{array}$$

Finally the chart of partials of f_3 is:



Algebraic Expansion Let us note that the function f is "of algebraic origin". That is, it is made up of three components and each component is a polynomial function in two variables⁵. Thus we can expect that simply plugging in sums and expanding out will be compatible with differential information. Let $(x, y) \in M$, $(h, k) \in \mathbb{R}^2$. We think of (x, y) as a position and (h, k) as a velocity or small perturbation. We'll plug in f the sum $(x + h, y + k)$, expand out the expressions and then organize the terms according to their orders in (h, k) :

⁵In fact, in the language we introduced the week after this problem set was due, we can say that f is a polynomial function.

$$\begin{aligned}
& f(x+h, y+k) \\
&= (f_1(x+h, y+k), f_2(x+h, y+k), f_3(x+h, y+k)) \\
&= ((x+h)(y+k) - 2(x+h) + (y+k) + 1, \\
&\quad (x+h)^2 + (y+k)^2 + (x+h) - 3(y+k) + 6, \\
&\quad (x+h)(y+k)^2) \\
&= (xy + xk + hy + hk - 2x - 2h + y + k + 1, \\
&\quad x^2 + 2xh + h^2 + y^2 + 2yk + k^2 + x + h - 3y - 3k + 6, \\
&\quad xy^2 + 2xyk + xk^2 + y^2h + 2yhk + hk^2) \\
&= \underbrace{(xy - 2x + y + 1, x^2 + y^2 + x - 3y + 6, xy^2)}_{\text{0th order}} \\
&\quad + \underbrace{((y-2)h + (x+1)k, (2x+1)h + (2y-3)k, y^2h + 2xyk)}_{\text{1st order}} \\
&\quad + \underbrace{(hk, h^2 + k^2, 2yhk + xk^2)}_{\text{2nd order}} \\
&\quad + \underbrace{(0, 0, hk^2)}_{\text{3rd order}}.
\end{aligned}$$

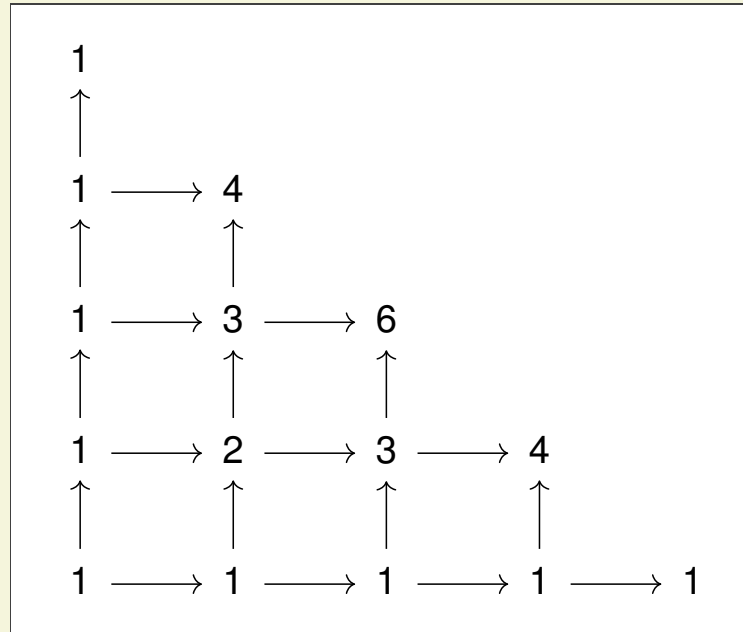
Here note that the 0th order term is exactly $f(x, y)$, that is, the 0th derivative of f at (x, y) . Similarly, the 1st order term ought to be related to the first derivative $f'(x, y)$ of f at (x, y) , the 2nd order term related to $f''(x, y)$ and 3rd order term to $f'''(x, y)$. The fact that there are no higher order terms in (h, k) ought to be related to the idea that all higher order derivatives of f are all zero.

Let us also observe the relation between the charts of partials we wrote above and the algebraic expansion. First focus on the tuple in the algebraic expansion that is of 1st order in (h, k) . In this tuple, the coefficients of h correspond exactly to the x -partials of

the components of f in the charts, and similarly the coefficients of k correspond exactly to the y -partials in the charts. In short, the variable $h = h^1 k^0$ corresponds to the $(1, 0)$ entries in the charts and the variable $k = h^0 k^1$ corresponds to the $(0, 1)$ entries in the charts. Note that this pattern applies to the 0th order term too: the variable $1 = h^0 k^0$ corresponds to the $(0, 0)$ entries in the charts.

Does the same pattern apply to higher order tuples in the algebraic expansion? Ultimately yes, but we'll need to reinterpret what the pattern actually is. Indeed, for instance focus on the tuple in the algebraic expansion that is of 2nd order in (h, k) . In this tuple, any component is made up of a linear combination of $h^2 = h^2 k^0$, $hk = h^1 k^1$ and $k^2 = h^0 k^2$. Again, it seems like the coefficients of $h^{\alpha_1} k^{\alpha_2}$ in the tuple correspond to the (α_1, α_2) entries in the charts, but at least for some coefficients there are multiplicative integer errors. I will leave it to you to how to reinterpret the pattern; for now it suffices to give two hints as to the cause of the multiplicative integer errors:

- (i) Much like in the single variable Taylor approximation, there should be correction factors: the k -th derivative ought to be corrected by $\frac{1}{k!}$ to match with the algebraic expansion.
- (ii) As the order of mixed partials ought not to matter, there should be correction factors due to different orders in which one can differentiate to match the algebraic expansion to the charts of partials; these correction factors should match with the **Pascal triangle** like so:



First Derivative To compute the first derivative of f' , we'll take the tuple in the algebraic expansion that is of first order in (h, k) , and write it as a linear function of (h, k) , for (x, y) fixed. Recall that our convention is that a linear transformation is encoded by a left multiplication by a rectangular matrix, so we'll write our tuples as column matrices. Thus

$$\begin{aligned}
 & \underbrace{((y-2)h + (x+1)k, (2x+1)h + (2y-3)k, y^2h + 2xyk)}_{\text{1st order}} \\
 &= \begin{pmatrix} (y-2)h + (x+1)k \\ (2x+1)h + (2y-3)k \\ y^2h + 2xyk \end{pmatrix} \\
 &= \underbrace{\begin{pmatrix} y-2 & x+1 \\ 2x+1 & 2y-3 \\ y^2 & 2xy \end{pmatrix}}_{=f'(x,y)} \begin{pmatrix} h \\ k \end{pmatrix} = f'(x, y)(h, k).
 \end{aligned}$$

Comparing with the charts of partials, we note that $f'(x, y)$ is the ma-

trix of first partials: row j corresponds to the component f_j , first column corresponds to the x -partials and the second column corresponds to the y -partials:

$$f'(x, y) = \begin{pmatrix} y - 2 & x + 1 \\ 2x + 1 & 2y - 3 \\ y^2 & 2xy \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x}(x, y) & \frac{\partial f_1}{\partial y}(x, y) \\ \frac{\partial f_2}{\partial x}(x, y) & \frac{\partial f_2}{\partial y}(x, y) \\ \frac{\partial f_3}{\partial x}(x, y) & \frac{\partial f_3}{\partial y}(x, y) \end{pmatrix}.$$

Finally let us do a **dimensional analysis**: according to our definition of the derivative, for each fixed $(x, y) \in M$, $f'(x, y)$ should be a linear transformation⁶ from $\mathbb{R}^2$ to $\mathbb{R}^3$, and hence the corresponding matrix should indeed be 3×2 .

Derivative at $(1, -1)$ Plugging in $(x, y) = (1, -1)$ in the 3×2 matrix corresponding to the derivative of f , we obtain

$$f'(1, -1) = \begin{pmatrix} -3 & 2 \\ 3 & -5 \\ 1 & -2 \end{pmatrix}.$$

Affine Approximation at $(1, -1)$ To compute the affine approximation of f at $(1, -1)$ we'll simply take the algebraic expansion, concatenate it by discarding the tuples corresponding to terms of order 2 or more, and then plugging in $(x, y) = (1, -1)$. Ultimately we'll obtain an affine function of (h, k) . In other words, from the point of view of calculus, the affine approximation⁷ at $(1, -1)$ is simply the function

⁶Recall that we proved in class that any linear transformation defined on a finite dimensional normed space is automatically bounded.

⁷Or, from the perspective of Taylor approximation, the **order 1 Taylor polynomial of f at $(1, -1)$**

$$T_1(f; (1, -1)) : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad (h, k) \mapsto f(1, -1) + f'(1, -1)(h, k).$$

Plugging $(x, y) = (1, -1)$ in $f(x, y)$, we obtain

$$\begin{aligned} f(1, -1) &= \underbrace{(xy - 2x + y + 1, x^2 + y^2 + x - 3y + 6, xy^2)}_{\text{0th order}}|_{(x,y)=(1,-1)} \\ &= (-1 - 2 - 1 + 1, 1 + 1 + 1 + 3 + 6, 1) \\ &= (-3, 12, 1). \end{aligned}$$

Thus in matrix notation the affine approximation of f at $(1, -1)$ is

$$T_1(f; (1, -1)) : \begin{pmatrix} h \\ k \end{pmatrix} \mapsto \begin{pmatrix} -3 \\ 12 \\ 1 \end{pmatrix} + \begin{pmatrix} -3 & 2 \\ 3 & -5 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix},$$

and in tuple notation we could write

$$T_1(f; (1, -1)) : (h, k) \mapsto (-3h + 2k - 3, 3h - 5k + 12, h - 2k + 1).$$

Directional Derivative at $(1, -1)$ in the Direction of $(15, 2)$ Since we know that f is differentiable everywhere, to compute its directional derivative at $(1, -1)$ in the direction of $(15, 2)$, we can simply compute $f'(1, -1)(15, 2)$:

$$f'(1, -1)(15, 2) = \begin{pmatrix} -3 & 2 \\ 3 & -5 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 15 \\ 2 \end{pmatrix} = \begin{pmatrix} -41 \\ 35 \\ 11 \end{pmatrix} = (-41, 35, 11).$$

We could have alternatively computed this directional derivative by taking a certain limit; I will leave the explicit calculation of the limit to you:

$$D_{(15,2)}f(1, -1) = \lim_{t \rightarrow 0} \frac{f(1 + 15t, -1 + 2t) - f(1, -1)}{t} = (-41, 35, 11).$$

Second Derivative Now we'll compute the second derivative of f , or more accurately the derivative of f' . Note that now we consider the function

$$f' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3), \quad (x, y) \mapsto \begin{pmatrix} y - 2 & x + 1 \\ 2x + 1 & 2y - 3 \\ y^2 & 2xy \end{pmatrix},$$

hence its derivative should be of the form

$$(f')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)).$$

Although f' takes values in a more complicated space compared to f , still it is of an "algebraic origin", so an algebraic expansion should suffice. Instead of using (h, k) we'll use $(\bar{h}, \bar{k})$ to denote the small perturbation:

$$\begin{aligned}
f'(x + \bar{h}, y + \bar{k}) &= \begin{pmatrix} (y + \bar{k}) - 2 & x + \bar{h} + 1 \\ 2(x + \bar{h}) + 1 & 2(y + \bar{k}) - 3 \\ (y + \bar{k})^2 & 2(x + \bar{h})(y + \bar{k}) \end{pmatrix} \\
&= \underbrace{\begin{pmatrix} y - 2 & x + 1 \\ 2x + 1 & 2y - 3 \\ y^2 & 2xy \end{pmatrix}}_{\text{0th order}} \\
&\quad + \underbrace{\begin{pmatrix} \bar{k} & \bar{h} \\ 2\bar{h} & 2\bar{k} \\ 2y\bar{k} & 2x\bar{k} + 2y\bar{h} \end{pmatrix}}_{\text{1st order}} \\
&\quad + \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ \bar{k}^2 & 2\bar{h}\bar{k} \end{pmatrix}}_{\text{2nd order}}.
\end{aligned}$$

Here we wrote the orders *from the perspective of* f' . As the algebraic expansion of f has nonzero terms up to third order, in a compatible way the algebraic expansion of f' has nonzero terms up to second order.

For fixed $(x, y) \in M$, the function that takes the tuple $(\bar{h}, \bar{k})$ to the 3×2 matrix that is the first order component in the algebraic expansion is a linear transformation, and hence is the derivative $(f')'(x, y)$:

$$(f')'(x, y) : \mathbb{R}^2 \rightarrow \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3), \quad (\bar{h}, \bar{k}) \mapsto \begin{pmatrix} \bar{k} & \bar{h} \\ 2\bar{h} & 2\bar{k} \\ 2y\bar{k} & 2x\bar{k} + 2y\bar{h} \end{pmatrix}.$$

Obtaining this expression is a perfectly valid computation of the second derivative $(f')'(x, y)$ of f at (x, y) .

We can not quite write a matrix that corresponds to this linear transformation like we did for $f'(x, y)$, at least without **flattening**. Still, we can keep using matrix algebra by being a little flexible as to what its rules are. Indeed, the linearity of the 3×2 matrix $(f')'(x, y)(\bar{h}, \bar{k})$ in $(\bar{h}, \bar{k})$ is equivalent to the linearity of each of its 6 entries, and hence we can consider $(f')'(x, y)$ to be a 3×2 matrix each of whose entries is a 1×2 matrix: for $(\bar{h}, \bar{k}) \in \mathbb{R}^2$, we have

$$\begin{aligned}
 (f')'(x, y)(\bar{h}, \bar{k}) &= \begin{pmatrix} \bar{k} & \bar{h} \\ 2\bar{h} & 2\bar{k} \\ 2y\bar{k} & 2x\bar{k} + 2y\bar{h} \end{pmatrix} \\
 &= \begin{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} & \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\ \begin{pmatrix} 2 & 0 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\ \begin{pmatrix} 0 & 2y \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} & \begin{pmatrix} 2y & 2x \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \end{pmatrix} \\
 &= \begin{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 2 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} \\ \begin{pmatrix} 0 & 2y \end{pmatrix} & \begin{pmatrix} 2y & 2x \end{pmatrix} \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\
 &= \underbrace{\begin{pmatrix} 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \\ 0 & 2y & 2y & 2x \end{pmatrix}}_{=(f')'(x, y)} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix}.
 \end{aligned}$$

Note that in the last expression above, it seems like we are multiplying a 3×4 matrix with a 2×1 matrix, and we want this product to

be a 3×2 matrix. Certainly this is not in accordance with the standard matrix algebra conventions! Still, insofar as we understand and keep in mind what this "matrix product" actually stands for, we can safely use it⁸.

Again note that the entries of $(f')'(x, y)$ correspond to the second order partials in the charts. and again it is no coincidence that some of the entries are repeated. I will leave it to you to establish the pattern, only giving two hints:

- (i) Row j corresponds to the component f_j ,
- (ii) Second and third columns are exactly the same.

Second Derivative via Multilinear Algebra Here is another way to think about $(f')'(x, y)$: For fixed (x, y) , $(\bar{h}, \bar{k}) \mapsto (f')'(x, y)(\bar{h}, \bar{k})$ is a linear function of linear functions from $\mathbb{R}^2$ to $\mathbb{R}^3$. That is to say, we have a bilinear function $\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^3$. So as to not confuse it with $(f')'(x, y)$, let us denote this bilinear function by $f''(x, y)$ ⁹:

$$f''(x, y) : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^3.$$

Thus we can feed two vectors (h, k) and $(\bar{h}, \bar{k})$ to it, and try to rearrange the resulting vector in $\mathbb{R}^3$. We'll use the key idea that to represent a scalar bilinear function we can let the matrix representation of $f''(x, y)$ to act from the left on one vector and from the right on the other vector:

⁸For instance addition and scalar multiplication work as usual, although not matrix multiplication. The formalism for these more complicated matrix products really is part of multilinear algebra, and it goes by names like "tensor product", "Kronecker product" or "outer product".

⁹As we discussed in class after the deadline for this problem set, this means that under the Currying isomorphism $\mathcal{L}_2(\mathbb{R}^2; \mathbb{R}^3) \cong \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3))$, $f''(x, y)$ transforms to $(f')'(x, y)$.

$$\begin{aligned}
f''(x, y)(\bar{h}, \bar{k})(h, k) &= \begin{pmatrix} \bar{k} & \bar{h} \\ 2\bar{h} & 2\bar{k} \\ 2y\bar{k} & 2x\bar{k} + 2y\bar{h} \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix} \\
&= \begin{pmatrix} \bar{k}h + \bar{h}k \\ 2\bar{h}h + 2\bar{k}k \\ 2y\bar{k}h + 2x\bar{k}k + 2y\bar{h}k \end{pmatrix} \\
&= \begin{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix}^T \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\ \begin{pmatrix} h \\ k \end{pmatrix}^T \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\ \begin{pmatrix} h \\ k \end{pmatrix}^T \begin{pmatrix} 0 & 2y \\ 2y & 2x \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \end{pmatrix} \\
&= \begin{pmatrix} h \\ k \end{pmatrix}^T \begin{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \\ \begin{pmatrix} 0 & 2y \\ 2y & 2x \end{pmatrix} \end{pmatrix} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix} \\
&= \begin{pmatrix} h \\ k \end{pmatrix}^T \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 2 & 0 \\ 0 & 2 \\ 0 & 2y \\ 2y & 2x \end{pmatrix}}_{=f''(x,y)} \begin{pmatrix} \bar{h} \\ \bar{k} \end{pmatrix}.
\end{aligned}$$

Again to make sense of the last equality we are forced to be more

flexible as to the rules of matrix algebra. Another important point is that the 2×2 entries of the 3×1 matrix representing $f''(x, y)$ are all **symmetric**; this indeed corresponds again to the fact that the order of mixed partials ought not matter.

Third Derivative The calculation of the third derivative of f , or more precisely the derivative of $(f')'$, is similar: Due to the algebraic nature of functions involved, an algebraic expansion should suffice. We start with the function

$$(f')' : M \rightarrow \mathcal{M}(3 \times 2; \mathcal{M}(1 \times 2; \mathbb{R})) \cong \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)),$$

$$(x, y) \mapsto \begin{pmatrix} 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \\ 0 & 2y & 2y & 2x \end{pmatrix}.$$

Let us use $(\tilde{h}, \tilde{k}) \in \mathbb{R}^2$ for the perturbation for the algebraic expansion:

$$\begin{aligned}
 (f')'(x + \tilde{h}, y + \tilde{k}) &= \begin{pmatrix} 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \\ 0 & 2(y + \tilde{k}) & 2(y + \tilde{k}) & 2(x + \tilde{h}) \end{pmatrix} \\
 &= \underbrace{\begin{pmatrix} 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 2 \\ 0 & 2y & 2y & 2x \end{pmatrix}}_{\text{0th order}} \\
 &\quad + \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2\tilde{k} & 2\tilde{k} & 2\tilde{h} \end{pmatrix}}_{\text{1st order}}.
 \end{aligned}$$

As before, the orders of the terms is from the perspective of $(f')'$; hence the k -th order term in $(f')'$ corresponds to the $(k + 1)$ -st order term in the expansion of f' , which in turn corresponds to the $(k + 2)$ -nd order term in the expansion of f .

We again note that the first order term in the algebraic expansion is a linear function of $(\tilde{h}, \tilde{k})$, and that linear function is $((f')')'(x, y)$. Relaxing the rules of matrix algebra, we can again obtain a matrix representation for the third derivative, this time obtaining a 3×2 matrix each of whose entries is a 1×2 matrix, with 1×2 matrix entries:

$$\begin{aligned}
((f')')'(x, y)(\tilde{h}, \tilde{k}) &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 2\tilde{k} & 2\tilde{k} & 2\tilde{h} \end{pmatrix} \\
&= \begin{pmatrix} \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} & \begin{pmatrix} 2 & 0 \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} \end{pmatrix} \\
&= \begin{pmatrix} \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} & \begin{pmatrix} 0 & 2 \end{pmatrix} & \begin{pmatrix} 2 & 0 \end{pmatrix} \end{pmatrix} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix} \\
&= \underbrace{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 2 & 2 & 0 \end{pmatrix}}_{=((f')')'(x, y)} \begin{pmatrix} \tilde{h} \\ \tilde{k} \end{pmatrix}.
\end{aligned}$$

Again I am leaving it to you to determine the patterns.

Third Derivative via Multilinear Algebra We could compute the second derivative two ways:

- (i) $(f')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; (\mathcal{L}(\mathbb{R}^2; \mathbb{R}^3))) \cong \mathcal{M}(3 \times 2; \mathcal{M}(1 \times 2; \mathbb{R}))$, and
- (ii) $f'' : M \rightarrow \mathcal{L}_2(\mathbb{R}^2; \mathbb{R}^3) \cong \mathcal{M}(3 \times 1; \mathcal{M}(2 \times 2; \mathbb{R}))$.

Similarly there are multiple ways to compute the third derivative:

- (i) $((f')')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)))$,

$$(ii) \ (f')'' : M \rightarrow \mathcal{L}_2(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3)),$$

$$(iii) \ (f'')' : M \rightarrow \mathcal{L}(\mathbb{R}^2; \mathcal{L}_2(\mathbb{R}^2; \mathbb{R}^3)),$$

$$(iv) \ f''' : M \rightarrow \mathcal{L}_3(\mathbb{R}^2; \mathbb{R}^3).$$

Above we really computed $(f')''$, by way of the isomorphism

$$\mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathcal{L}(\mathbb{R}^2; \mathbb{R}^3))) \cong \mathcal{M}(3 \times 2; \mathcal{M}(1 \times 2; \mathcal{M}(1 \times 2; \mathbb{R}))).$$

Ultimately all of these functions ought to be compatible; I am leaving it to you to actually compute these different looking third derivatives; for instance for the third option you would differentiate the following function:

$$f'' : M \rightarrow \mathcal{M}(3 \times 1; \mathcal{M}(2 \times 2; \mathbb{R})) \cong \mathcal{L}_2(\mathbb{R}^2; \mathbb{R}^3),$$

$$(x, y) \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 2 & 0 \\ 0 & 2 \\ 0 & 2y \\ 2y & 2x \end{pmatrix}.$$

Fair warning: flexing the rules of matrix algebra to encode trilinearity for f''' will likely prove to be harder than the other computations.

Follow-up Exercise 1 Find and correct all typos and computation errors in the solution above.

Follow-up Exercise 2 Complete all calculations left to you in the solution above.

Follow-up Exercise 3 Redo all calculations in the solution above by flattening the objects involved into column matrices at each step.

6 PSet 6, Problem 2

Let M be an open subset of $\mathbb{R}^2$, $f \in C^1(M; \mathbb{R})$. Then there is a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ such that for any $(x, y) \in M : f(x, y) = \phi(x)$ iff for any $(x, y) \in M$, $\frac{\partial f}{\partial y}(x, y) = 0$.

The given statement in general works only in one direction, but with a crucial extra assumption one can recover equivalence.

Claim: Let M be an open subset of $\mathbb{R}^2$, $f \in C^1(M; \mathbb{R})$. If there is a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ such that for any $(x, y) \in M : f(x, y) = \phi(x)$, then for any $(x, y) \in M$, $\frac{\partial f}{\partial y}(x, y) = 0$.

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Proof: Let $(x, y) \in M$. Using the limit characterization of the y -partial of f , we have:

$$\begin{aligned} \frac{\partial f}{\partial y}(x, y) &= \lim_{k \rightarrow 0} \frac{f(x, y + k) - f(x, y)}{k} \\ &= \lim_{k \rightarrow 0} \frac{\phi(x) - \phi(x)}{k} \\ &= 0. \end{aligned}$$

Note that here k is a scalar.

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Claim: There are open subsets $M \subseteq \mathbb{R}^2$ and some $f \in C^1(M; \mathbb{R})$ with the following properties:

- (i) f is not constant in y , that is, there is no function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ such that for any $(x, y) \in M : f(x, y) = \phi(x)$, and
- (ii) for any $(x, y) \in M$, $\frac{\partial f}{\partial y}(x, y) = 0$.

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Proof: For instance consider the open annulus

$$M = \mathbb{R}^2[0, 1[\cup]1, 2] = \mathbb{R}^2[0, 2] \setminus \mathbb{R}^2[0, 1].$$

Let us first verify that M is indeed an open subset of the plane. Let $(x, y) \in M$. Thus by definition

$$1 \leq |(x, y)| = \sqrt{x^2 + y^2} \leq 2.$$

Put $\delta = \min\{|(x, y)| - 1, 2 - |(x, y)|\}$ and let $(\bar{x}, \bar{y}) \in \mathbb{R}^2[|(x, y)| \leq \delta]$. We claim that $(\bar{x}, \bar{y}) \in M$. Indeed,

$$|(\bar{x}, \bar{y})| - |(x, y)| \leq ||(\bar{x}, \bar{y})| - |(x, y)|| \leq |(\bar{x}, \bar{y}) - (x, y)| \leq \delta \leq 2 - |(x, y)|$$

implies that $|(\bar{x}, \bar{y})| \leq 2$, and similarly

$$|(x, y)| - |(\bar{x}, \bar{y})| \leq ||(\bar{x}, \bar{y})| - |(x, y)|| \leq |(\bar{x}, \bar{y}) - (x, y)| \leq \delta \leq |(x, y)| - 1$$

implies that $|(\bar{x}, \bar{y})| \geq 1$, so that $(\bar{x}, \bar{y}) \in M$, and hence M is open.

Now define $f : M \rightarrow \mathbb{R}$ by

$$(x, y) \mapsto \begin{cases} b(x), & \text{if } y \geq 0 \\ -b(x), & \text{if } y \leq 0 \end{cases},$$

where $b : x \mapsto e^{1/(x^2-1)}$ is a **bump function** with the property that for $|x| \geq 1$, $b(x) = 0$. Using methods from MATH 3210¹⁰ one obtains that $b \in C^\infty$ (and in particular $b \in C^1$). A further straightforward calculation shows that for any $(x, y) \in M$,

¹⁰See e.g. problem 6 in **problem set 6** from last semester's MATH 3210.

$$\frac{\partial f}{\partial x}(x, y) = b'(x), \quad \frac{\partial f}{\partial y}(x, y) = 0,$$

hence by a theorem proved in class, $f \in C^\infty(M; \mathbb{R})$. Finally this function is not constant in y , since

$$f(0, 1) = b(0) = e^{-1}; \quad f(0, -1) = -b(0) = -e^{-1}.$$

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Note that in the example we constructed above, the open subset M is path-connected. Hence even on path-connected open subsets, the statement fails. Under the more restrictive assumption that M is **convex** however, one obtains a result in the affirmative. A subset S of a vector space V is **convex** if for any $v_1, v_2 \in S$ and for any $\tau \in [0, 1]$, one also has $(1 - \tau)v_1 + \tau v_2 \in S$. In words, S is **convex** if for any two points in S , the line segment connecting the two points is completely within S .

Claim: Let M be an open subset of $\mathbb{R}^2$, $f \in C^1(M; \mathbb{R})$. If M is convex, and for any $(x, y) \in M$, $\frac{\partial f}{\partial y}(x, y) = 0$, then there is a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ such that for any $(x, y) \in M$: $f(x, y) = \phi(x)$.

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Proof: Let $(x, y_1), (x, y_2) \in M$. We claim that $f(x, y_1) = f(x, y_2)$, that is, f is constant in the second entry. By the convexity assumption, for any $t \in [0, 1]$, $(x, (1 - t)y_1 + ty_2) \in M$. Thus we may apply the Mean Value Theorem to f between these two points to obtain

$$|f(x, y_1) - f(x, y_2)| \leq \left(\sup_{t \in [0, 1]} \left| \frac{\partial f}{\partial y}(x, (1 - t)y_1 + ty_2) \right| \right) |y_2 - y_1| = 0.$$

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Follow-up Exercise 1 Plot the example given in the above solution in a 3D graphing calculator. **There is nothing to prove in this problem.**

Follow-up Exercise 2 Call a subset $S \subseteq \mathbb{R}^2$ **vertically convex** if for any $(x_1, y_1), (x_2, y_2) \in S$, if $x_1 = x_2$, then the line segment connecting (x_1, y_1) and (x_2, y_2) is completely inside S .

- (i) A subset $S \subseteq \mathbb{R}^2$ is convex iff it is vertically convex.
- (ii) Let M be an open vertically convex subset of $\mathbb{R}^2$, $f \in C^1(M; \mathbb{R})$. Then there is a function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ such that for any $(x, y) \in M$:
 $f(x, y) = \phi(x)$ iff for any $(x, y) \in M$, $\frac{\partial f}{\partial y}(x, y) = 0$.
- (iii) Let M be an open subset of $\mathbb{R}^2$. Then M is vertically convex iff for any $f \in C^1(M; \mathbb{R})$,

$$\begin{aligned} & \exists \phi : \mathbb{R} \rightarrow \mathbb{R}, \forall (x, y) \in M : f(x, y) = \phi(x) \\ & \Leftrightarrow \forall (x, y) \in M : \frac{\partial f}{\partial y}(x, y) = 0. \end{aligned}$$

Follow-up Exercise 3 Let E, F be normed spaces, $M \subseteq E$, $N \subseteq F$ be open, $E = E_1 \times E_2$, $f : M \rightarrow N$ be C^1 .

- (i) M is convex iff $M = M_1 \times M_2$ for some open subsets $M_1 \subseteq E_1$, $M_2 \subseteq E_2$.
- (ii) If M is convex, then for any $(x_1, y_1), (x_2, y_2) \in M$, $x_1 = x_2$ implies $f(x_1, y_1) = f(x_2, y_2)$ iff $D_2 f = 0$.

7 PSet 7, Problem 3

Let E and F be normed spaces, $M \subseteq E$ and $N \subseteq F$ be nonempty open subsets, and $p : M \rightarrow N$ be a polynomial function.

- (i) If $T_\bullet$ and $S_\bullet$ are both sequences of bounded multilinear transformations associated to p , then for any nonnegative integer n , $T_n = 0$ iff $S_n = 0$.
- (ii) The degree of p is a well-defined nonnegative integer.
- (iii) There is a unique sequence of bounded multilinear transformations associated to p .
- (iv) There is a unique sequence of bounded symmetric multilinear transformations associated to p .
- (v) For any nonnegative integer k , the k -th derivative $p^{(k)}$ of p is a polynomial function.
- (vi) $p \in C^\infty(M; N)$.

A Note This problem was initially chosen as the problem to be graded for accuracy for pset 7, however upon further consideration, due to the complexity of index management it too was graded for completion. Still it is instructive to go through some examples to convince oneself that indeed even in the multivariable setting polynomial functions are the "nice functions" for the purposes of differential calculus.

It is safe to assume that you will not be asked to derive or memorize formulas derived below, for examination purposes. However you might be asked to compute the derivatives of low order of polynomials of

low degree.

The Single Variable Case Let us first discuss the single variable case to guide our intuition. Thus we take $E = F = \mathbb{R}$. In this case, a k -linear function $T_k : E^k \rightarrow F$ would be automatically bounded and further it would be completely determined by one real number, namely $a_k = T_k(1, 1, \dots, 1) \in F$, as for any $v_1, v_2, \dots, v_k \in E$,

$$T_k(v_1, v_2, \dots, v_k) = a_k \prod_{i=1}^k v_i.$$

Since multiplication of numbers is symmetric, any k -linear function would be automatically symmetric.

Thus a polynomial function $p : M \rightarrow N$ would be so that for some sequence of real numbers a_n such that $a_n = 0$ for all but finitely many integers n , and for any $x \in M$,

$$p(x) = \sum_{n=0}^{\infty} a_n x^n.$$

Consider the set of nonnegative integers $\{n \mid \forall m \geq n : a_m = 0\}$. This is a nonempty set that is bounded from below, hence by the well-ordering principle it has a minimum; let us denote that minimum by d . Thus we may write

$$\forall x \in M : p(x) = \sum_{k=0}^d a_k x^k.$$

Let us compute the derivatives of p . As p is a linear combination of multilinear functions, it is helpful to first record the derivatives of monomials. We have:

$$\forall k, \ell \in \mathbb{Z}_{\geq 0}, \forall x \in E : \frac{d^\ell x^k}{dx^\ell}(x) = \begin{cases} \frac{k!}{(k-\ell)!} x^{k-\ell} & , \text{ if } \ell \leq k \\ k! & , \text{ if } \ell = k \\ 0 & , \text{ if } \ell \geq k, \end{cases}$$

and consequently,

$$\forall \ell \in \mathbb{Z}_{\geq 0}, \forall x \in M : p^{(\ell)}(x) = \begin{cases} \sum_{k=\ell}^d a_k \frac{k!}{(k-\ell)!} x^{k-\ell} & , \text{ if } \ell \leq d \\ a_d d! & , \text{ if } \ell = d \\ 0 & , \text{ if } \ell \geq d. \end{cases}$$

In particular $p \in C^\infty(M; N)$ and any derivative of p itself is a polynomial function.

We now need to address the questions around the multilinear expansion of p . Suppose there is another sequence $b_\bullet$ of real numbers such that $b_n = 0$ for all but finitely many integers n and for any $x \in M$,

$$p(x) = \sum_{n=0}^{\infty} b_n x^n.$$

Without loss of generality, we may assume that for any $n \geq d$, $b_n = 0$. Consider the function $\phi : M \rightarrow F$ defined by

$$\phi : x \mapsto \sum_n b_n x^n - \sum_n a_n x^n = \sum_{k=0}^d (b_k - a_k) x^k.$$

Note that $\phi(x) = 0$ for any $x \in M$; hence any of its derivatives at any $x \in M$ is zero. On the other hand, ϕ in and of itself is a polynomial function, so applying our formula for the order $\ell = d$ derivative of a polynomial we obtain, using any point $x \in M$,

$$0 = \phi^{(d)}(x) = (b_d - a_d)d!.$$

The factorial of no nonnegative integer is zero, hence $b_d = a_d$, and

$$\phi : x \mapsto \sum_{k=0}^{d-1} (b_k - a_k)x^k.$$

Now using the formula for the order $\ell = d - 1$ derivative, similarly we obtain $b_{d-1} = a_{d-1}$. By the Principle of Mathematical Induction¹¹, for any n , $a_n = b_n$. This finishes the discussion in the single variable case; to summarize we confirmed all the claims made in the problem.

The General Case

The Line of Attack The single variable case ought to be our guide for the general case. Just like the single variable case, instead of verifying the claims in the order they are given, our line of attack will be as follows:

- (i) Derive a formula for the ℓ -th derivative of a bounded k -linear function $E^k \rightarrow F$.
- (ii) Derive a formula for the ℓ -th derivative of a polynomial function $p : M \rightarrow N$.
- (iii) Having a formula for all derivatives will allow us to establish that any derivative of a polynomial function is itself a polynomial function and consequently also that $p \in C^\infty$.
- (iv) If we have two families of bounded multilinear functions associated to the same polynomial function, we'll consider their

¹¹I am leaving it to you to write this more formally; technically here we don't need to evoke PMI as there are only finitely many steps we need to consider.

difference. As spaces of bounded multilinear maps are vector spaces, this difference itself is a polynomial function, and it is constantly zero on M . We'll then index into each of the summands and study each order separately.

Collapse Operators Recall that for "functions of algebraic origin", computing derivatives ought to be closely related to an algebraic expansion. Thus for instance for operator composition $\mathcal{C} : (A_1, A_2) \mapsto A_1 \circ A_2$, which you have verified in [problem set 1](#) to be bilinear and in [problem set 2](#) to be bounded bilinear, we have

$$\begin{aligned}\mathcal{C}(A_1 + H_1, A_2 + H_2) &= (A_1 + H_1)(A_2 + H_2) \\ &= A_1 A_2 + A_1 H_2 + H_1 A_2 + H_1 H_2 \\ &= \mathcal{C}(A_1, A_2) + \mathcal{C}(A_1, H_2) + \mathcal{C}(H_1, A_2) + \mathcal{C}(H_1, H_2),\end{aligned}$$

hence by the uniqueness of Taylor approximation,

$$\mathcal{C}^{(\ell)}(A_1, A_2)(H_1, H_2)^\ell = \begin{cases} \mathcal{C}(A_1, A_2) & , \text{ if } \ell = 0 \\ \mathcal{C}(A_1, H_2) + \mathcal{C}(H_1, A_2) & , \text{ if } \ell = 1 \\ 2\mathcal{C}(H_1, H_2) & , \text{ if } \ell = 2 \\ 0 & , \text{ if } \ell \geq 3 \end{cases},$$

and $\mathcal{C} \in C^\infty$. Note that in this formula we don't quite have an expression for the second derivative; after all, the second derivative $\mathcal{C}''(A_1, A_2)$ itself ought to be a bounded bilinear function in and of itself, thus we should really look for a formula for

$$\mathcal{C}''(A_1, A_2)((H_1, H_2), (K_1, K_2))$$

that is symmetric, so that

$$\mathcal{C}''(A_1, A_2)((H_1, H_2), (K_1, K_2)) = \mathcal{C}''(A_1, A_2)((K_1, K_2), (H_1, H_2)),$$

and that matches with the above formula if $(H_1, H_2) = (K_1, K_2)$. I'll leave it to you to verify that (say by computing $(\mathcal{C}')'$ first and then reshaping it to $\mathcal{C}''$ by unCurrying), the true formula for the second derivative here is

$$\mathcal{C}''(A_1, A_2)((H_1, H_2), (K_1, K_2)) = \mathcal{C}(H_1, K_2) + \mathcal{C}(K_1, H_2).$$

While such expressions are relatively easy to manage for multilinear functions that take few vectors as input, you can imagine the expressions for derivatives get more complicated for a bounded k -linear as k increases. As the example of operator composition shows, the main logistical challenge is to keep track of into which arguments of the multilinear function we are considering as a derivative we should plug in the components of the input vectors. Thus being more systematic regarding the evaluation of multilinear functions will be very useful. For this purpose we'll introduce "collapse operators" as follows:

Definition 1: Let E and F be normed spaces, $k \in \mathbb{Z}_{\geq 1}$. For any $v \in E$, any $T \in \mathcal{BL}_k(E; F)$ and any $i \in \{1, 2, \dots, k\}$, we define the **collapse**¹² $v \lrcorner_i T \in \mathcal{BL}_{k-1}(E; F)$ of T by v at the i -th argument by

$$v \lrcorner_i T : (w_1, w_2, \dots, w_{k-1}) \mapsto T(w_1, w_2, \dots, w_{i-1}, v, w_i, \dots, w_{k-1}).$$

¹²More commonly this operation is called **tensor contraction**, at least for symmetric multilinear functions between finite dimensional normed spaces. As we are not using the tensor formalism in this class, and we use "contraction" for a **completely different notion**, I decided to use a different name here.

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We have the following lemma governing the collapse transformations:

Lemma 2: Let E and F be normed spaces, $k \in \mathbb{Z}_{\geq 1}$. Then for any $i \in \{1, 2, \dots, k\}$,

$$\bullet \lrcorner_i \bullet : E \times \mathcal{BL}_k(E; F) \rightarrow \mathcal{BL}_{k-1}(E; F), \\ (v, T) \mapsto v \lrcorner_i T$$

is a bounded bilinear operator.

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The proof of this lemma involves not more than reinterpreting some of the bounds and definitions we have been using throughout the semester; I am leaving the proof to you.

A 0-linear function $E \rightarrow F$ is nothing but a vector in F , and so if $v \in E$, as vectors in E in general should have nothing to do with vectors in F , the natural choice for v to collapse a 0-linear function is the zero transformation. Similarly, if for a k -linear function, the i -th argument is collapsed once, any subsequent collapse of the i -th argument ought to transform the original k -linear function to the zero vector in $\mathcal{BL}_{k-1}(E; F)$. Based on this, if $v, w \in E$, and $i, j \in \{1, 2, \dots, k\}$ with $i \neq j$, we put

$$v \lrcorner_i w \lrcorner_j = 0, \quad (v, w) \lrcorner_{ij} = v \lrcorner_i w \lrcorner_j = w \lrcorner_j v \lrcorner_i.$$

We similarly abbreviate the composition of an arbitrary but finite number of collapse operators in different arguments as the collapse by a tuple of vectors. The benefit of using the collapse operators is to help us switch from thinking multilinear functions acting on vectors to

vectors acting on multilinear functions.

Going back to our operator composition example, here is how one can rewrite the expressions for the derivatives using collapse operators:

$$\begin{aligned}
 \mathcal{C}^{(0)}(A_1, A_2) &= \mathcal{C}(A_1, A_2) \\
 &= (A_1, A_2) \lrcorner_{12} \mathcal{C} \\
 &= A_1 \lrcorner_1 A_2 \lrcorner_2 \mathcal{C} \\
 &= A_2 \lrcorner_2 A_1 \lrcorner_1 \mathcal{C}, \\
 \mathcal{C}'(A_1, A_2)(H_1, H_2) &= \mathcal{C}(A_1, H_2) + \mathcal{C}(H_1, A_2) \\
 &= [(A_1, H_2) \lrcorner_{12} + (H_1, A_2) \lrcorner_{12}] \mathcal{C} \\
 &= (H_1 \lrcorner_1 + H_2 \lrcorner_2)(A_1 \lrcorner_1 + A_2 \lrcorner_2) \mathcal{C} \\
 &= [(A_1 \lrcorner_1 + A_2 \lrcorner_2) \mathcal{C}](H_1, H_2), \\
 \mathcal{C}''(A_1, A_2)((H_1, H_2), (K_1, K_2)) &= \mathcal{C}(H_1, K_2) + \mathcal{C}(K_1, H_2) \\
 &= [(H_1, K_2) \lrcorner_{12} + (K_1, H_2) \lrcorner_{12}] \mathcal{C} \\
 &= (K_1 \lrcorner_1 + K_2 \lrcorner_2)(H_1 \lrcorner_1 + H_2 \lrcorner_2) \mathcal{C}.
 \end{aligned}$$

It looks like collapse operators make the formulas more complicated, but for k large, we'll be able to write formulas for derivatives of bounded k -linears as finite compositions of finite sums of collapse operators, appropriately indexed.

Derivatives of Bounded Multilinear Functions Now that we defined the new collapse notation, we can write a formula for derivatives of bounded multilinear operators. I'll again leave it to you to verify that the formula below is correct (say by way of induction). Let $k \in \mathbb{Z}_{\geq 0}$, $T_k : E^k \rightarrow F$ be bounded k -linear, $x = (x_1, x_2, \dots, x_k) \in E^k$, $\ell \in \mathbb{Z}_{\geq 0}$. Then the ℓ -th derivative $T_k^{(\ell)}(x)$ of T_k at x ought to be a

bounded ℓ -linear operator

$$T_k^{(\ell)}(x) : \underbrace{E^k \times E^k \times \dots \times E^k}_{\ell \text{ many}} \rightarrow F.$$

Let $h_1, h_2, \dots, h_\ell \in E^k$ be arbitrary, so that we have ℓ many k -tuples of vectors in E . Put

$$\begin{aligned} h_1 &= (h_{11}, h_{12}, \dots, h_{1k}) \\ h_2 &= (h_{21}, h_{22}, \dots, h_{2k}) \\ &\vdots \\ h_\ell &= (h_{\ell 1}, h_{\ell 2}, \dots, h_{\ell k}); \end{aligned}$$

here each h_{ij} is one vector in E . Then if $\ell \leq k$, one has

$$\begin{aligned} &T_k^{(\ell)}(x)(h_1, h_2, \dots, h_\ell) \\ &= \left(\sum_{i_\ell=1}^k h_{\ell i_\ell} \cdot i_{i_\ell} \right) \cdots \left(\sum_{i_2=1}^k h_{2 i_2} \cdot i_{i_2} \right) \left(\sum_{i_1=1}^k h_{1 i_1} \cdot i_{i_1} \right) \\ &\quad \left(\sum_{1 \leq j_1 \leq j_2 \leq \dots \leq j_\ell \leq k} x_{\widehat{j_1 j_2 \dots j_\ell}} \cdot \widehat{i_{j_1 j_2 \dots j_\ell}} \right) T_k, \end{aligned}$$

where $x_{\widehat{j_1 j_2 \dots j_\ell}} \in E^{k-\ell}$ is the tuple of vectors one obtains by dropping the j_m components from x , and $\widehat{j_1 j_2 \dots j_\ell}$ is precisely the associated remaining indices. Note further that

$$\left(\sum_{1 \leq j_1 \leq j_2 \leq \dots \leq j_\ell \leq k} x_{\widehat{j_1 j_2 \dots j_\ell}} \cdot \widehat{i_{j_1 j_2 \dots j_\ell}} \right) T_k$$

is by the nature of collapse operators a bounded ℓ -linear transformation (as we are collapsing $k - \ell$ arguments of a k -linear function). Similarly for $\ell = k$ the factor that is the linear combination of collapse operators coming from components of x completely drops out and we obtain

$$\begin{aligned}
 T_k^{(k)}(x)(h_1, h_2, \dots, h_k) &= \left(\sum_{i_k=1}^k h_{ki_k} \lrcorner_{i_k} \right) \cdots \left(\sum_{i_2=1}^k h_{2i_2} \lrcorner_{i_2} \right) \left(\sum_{i_1=1}^k h_{1i_1} \lrcorner_{i_1} \right) T_k \\
 &= \sum_{\sigma \in \text{Bij}_k} (h_{\sigma(1)1}, h_{\sigma(2)2}, \dots, h_{\sigma(k)k}) \lrcorner T_k \\
 &= \sum_{\sigma \in \text{Bij}_k} T_k(h_{\sigma(1)1}, h_{\sigma(2)2}, \dots, h_{\sigma(k)k}),
 \end{aligned}$$

where Bij_k is the collection of all **permutations** of the set $\{1, 2, \dots, k\}$. Note also that by this formula we are guaranteed that $T_k^{(k)}(x)$ is a symmetric bounded k -linear operator. Further, $T_k^{(k)}(x)$ is constant in x , and hence for $\ell \geq k$, $T_k^{(\ell)}(x) = 0$. We can summarize all these formulas as follows, suppressing the indices for collapse operators, with the understanding that the vectors ought to be used to collapse T_k in the appropriate arguments:

$$T_k^{(\ell)}(x)(h_1, h_2, \dots, h_\ell) = \begin{cases} \left(\sum_{i_\ell=1}^k h_{\ell i_\ell} \lrcorner \right) \cdots \left(\sum_{i_1=1}^k h_{1 i_1} \lrcorner \right) \left(\sum_J x_J \lrcorner \right) T_k & , \text{ if } \ell < k \\ \sum_{\sigma \in \text{Bij}_k} T_k(h_{\sigma(1)1}, h_{\sigma(2)2}, \dots, h_{\sigma(k)k}) & , \text{ if } \ell = k \\ 0 & , \text{ if } \ell > k, \end{cases}$$

Here $J = (j_1, j_2, \dots, j_\ell)$ is an arbitrary tuple of integers such that

$1 \leq j_1 \leq \dots \leq j_\ell \leq k$ and $x_{\hat{J} \downarrow}$ is the collapse operator obtained by dropping all the components of x with indices in J . Consequently $\left(\sum_J x_{\hat{J} \downarrow}\right) T_k$ is bounded ℓ -linear.

Derivatives of Compositions We computed the derivatives of bounded multilinear. To derive the formula for derivatives of a polynomial function we'll need to compute the derivatives of composition of two functions. Indeed, as

$$p : x \mapsto \sum_n T_n(x, x, \dots, x),$$

we really need formulas for derivatives of functions of the form $T_n \circ \text{diag}_n$, where T_n is bounded n -linear and $\text{diag}_n : E \rightarrow E^n$ is the function defined by $x \mapsto (x, x, \dots, x)$. There is a **general formula** for the derivatives of the composition of sufficiently many differentiable functions, although in this case since the inner function diag_n is bounded linear, derivation of a formula requires significantly less logistics:

Lemma 3: Let V, W, X be normed spaces, $A : V \rightarrow W$ be bounded linear, $f : W \rightarrow X$ be C^∞ . Then for any $\ell \in \mathbb{Z}_{\geq 0}$, for any $x \in V$ and any $h_1, h_2, \dots, h_\ell \in V$:

$$(f \circ A)^{(\ell)}(x)(h_1, h_2, \dots, h_\ell) = f^{(\ell)} \circ A(x)(Ah_1, Ah_2, \dots, Ah_\ell).$$

┘

Again I am leaving it to you to establish this lemma; one method would be to simply to expand $f \circ A(x + h)$ up to order ℓ using Taylor approximation and the linearity of A .

This lemma in particular implies that for any $k \in \mathbb{Z}_{\geq 0}$, for any bounded k -linear $T_k : E^k \rightarrow F$, for any $x \in E$ and for any $h_1, h_2, \dots, h_\ell \in$

E:

$$\begin{aligned}
& (T_k \circ \text{diag}_k)^{(\ell)}(x)(h_1, h_2, \dots, h_\ell) \\
&= \begin{cases} \left(\sum_{i_\ell=1}^k h_\ell \lrcorner_{i_\ell} \right) \cdots \left(\sum_{i_1=1}^k h_1 \lrcorner_{i_1} \right) \left(\sum_J \text{diag}_k(x) \hat{\lrcorner}_J \right) T_k & , \text{ if } \ell \leq k \\ \sum_{\sigma \in \text{Bij}_k} T_k(h_{\sigma(1)}, h_{\sigma(2)}, \dots, h_{\sigma(k)}) & , \text{ if } \ell = k \\ 0 & , \text{ if } \ell \geq k. \end{cases}
\end{aligned}$$

Derivatives of Polynomial Functions We can now focus on polynomial functions. Let $p : M \rightarrow N$ be a polynomial function and let $T_\bullet$ be a sequence of multilinear functions associated to p . As in the single variable case, denote by $d \in \mathbb{Z}_{\geq 0}$ the least integer such that for any $m \geq d$, $T_m = 0$, so that we have:

$$\forall x \in M : p(x) = \sum_{k=0}^d T_k(\underbrace{x, x, \dots, x}_{k \text{ many}}) = \sum_{k=0}^d T_k \circ \text{diag}_k(x).$$

Note that at this stage, the integer d depends on $T_\bullet$, even though ultimately we'll show that it is completely determined by the polynomial function p itself.

Let $\ell \in \mathbb{Z}_{\geq 0}$. Then the ℓ -th derivative $p^{(\ell)}$ of p is by definition a function

$$p^{(\ell)} : M \rightarrow \mathcal{BL}_\ell(E; F).$$

Our aim is to show that *this* function itself is a polynomial function. So we need to associate to this function a family of bounded multilinear functions that transform tuples of vectors in E into vectors in the normed space $\mathcal{BL}_\ell(E; F)$. Let $x \in M$. By the linearity of derivatives and the derivative formulas for bounded multilinear, if $\ell \leq d$, we have that $p \in C^\ell(M; N)$ and for any $h_1, h_2, \dots, h_\ell \in E$:

$$\begin{aligned}
& p^{(\ell)}(x)(h_1, h_2, \dots, h_\ell) \\
&= \sum_{k=0}^d (T_k \circ \text{diag}_k)^{(\ell)}(x)(h_1, h_2, \dots, h_\ell) \\
&= \sum_{k=0}^{\ell-1} \underbrace{(T_k \circ \text{diag}_k)^{(\ell)}(x)(h_1, h_2, \dots, h_\ell)}_{=0} \\
&\quad + \underbrace{(T_\ell \circ \text{diag}_\ell)^{(\ell)}(x)(h_1, h_2, \dots, h_\ell)}_{\in \mathcal{BS}_\ell(E;F)} \\
&\quad + \sum_{k=\ell+1}^d \underbrace{(T_k \circ \text{diag}_k)^{(\ell)}(x)(h_1, h_2, \dots, h_\ell)}_{\in \mathcal{BL}_\ell(E;F)} \\
&= (T_\ell \circ \text{diag}_\ell)^{(\ell)}(x)(h_1, h_2, \dots, h_\ell) \\
&\quad + \sum_{k=\ell+1}^d (T_k \circ \text{diag}_k)^{(\ell)}(x)(h_1, h_2, \dots, h_\ell) \\
&= \sum_{k=\ell}^d \left(\sum_{i_\ell=1}^k h_{\ell \dashv i_\ell} \right) \cdots \left(\sum_{i_1=1}^k h_{1 \dashv i_1} \right) \left(\sum_J \text{diag}_k(x)_{\widehat{J} \dashv} \right) T_k.
\end{aligned}$$

We had earlier established that for any k , and for any J ,

$$x_{\widehat{J}} \mapsto x_{\widehat{J} \dashv} T_k$$

is bounded $(k - \ell)$ -linear from $E^{k-\ell}$ to $\mathcal{BL}_\ell(E; F)$, and in the latest formula we have a finite sum of such ℓ -linear valued $(k - \ell)$ -linears, where in each argument the same point $x \in M$ is used. This justifies the claim that $p^{(\ell)}$ is a polynomial function. Similarly we obtain for $\ell = d$, that

$$p^{(d)}(x) = (T_d \circ \text{diag}_d)^{(d)}(x) \in \mathcal{BS}_d(E; F),$$

and for $\ell \neq d$, that $p^{(\ell)} = 0$. Consequently, any derivative of p is itself a polynomial function and $p \in C^\infty(M; N)$.

Symmetry Considerations In the general case, multilinear functions are not completely determined by a unique number, and we'll need to be careful about the symmetry property. Even in the finite dimensional case, there are multiple sequences associated to a given polynomial function; for instance consider the polynomial

$$\mathbb{R}^2 \rightarrow \mathbb{R}, (x, y) \mapsto xy.$$

For any $\alpha \in \mathbb{R}$, consider the bilinear function

$$T_{2,\alpha} : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R},$$

$$((x_1, y_1), (x_2, y_2)) \mapsto \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}^T \begin{pmatrix} 0 & \alpha \\ 1 - \alpha & 0 \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}.$$

For $\alpha \neq 1/2$, $T_{2,\alpha}$ is not symmetric, but for any $\alpha \in \mathbb{R}$ and for any $(x, y) \in \mathbb{R}^2$, $xy = T_{2,\alpha}(x, y)^2$.

Existence In general too one can not hope to associate a unique sequence of bounded multilinear functions to a given polynomial function. However one can indeed associate a unique sequence of *symmetric* bounded multilinear functions. The formula for the k -th derivative of a bounded k -linear is symmetric, and based on this one can define a **symmetrization** operator on bounded k -linear operators:

$$\text{Sym}_k : \mathcal{BL}_k(E; F) \rightarrow \mathcal{BS}_k(E; F),$$

$$T_k \mapsto \left[(h_1, h_2, \dots, h_k) \mapsto \frac{1}{k!} \sum_{\sigma \in \text{Bij}_k} T_k(h_{\sigma(1)}, h_{\sigma(2)}, \dots, h_{\sigma(k)}) \right].$$

I'll leave it to you to establish linear algebraic properties of the symmetrization operator (e.g. it is bounded linear and idempotent, its image is $\mathcal{BS}_k(E; F)$ and its kernel is $\mathcal{BA}_k(E; F)$), as well as the fact that if $T_\bullet$ is a sequence of bounded multilinear functions associated to a polynomial, then the symmetrized sequence $\text{Sym}_\bullet(T_\bullet)$ too is associated to the same polynomial. Another important observation will be that for any $x \in M$ and for any $k \in \mathbb{Z}_{\geq 0}$:

$$k! \text{Sym}_k(T_k) = (T_k \circ \text{diag}_k)^{(k)}(x).$$

Uniqueness Finally we establish the uniqueness of a sequence of *symmetric* bounded multilinear functions associated to p . By symmetrizing $T_\bullet$ if needed, we assume that $T_\bullet$ is a sequence of symmetric bounded multilinear functions associated to p . Suppose now $S_\bullet$ is another sequence of symmetric bounded multilinear functions associated to p . As in the single variable case, we may assume that for any $n \geq d$, that $S_n = 0$. Consider the function

$$\phi : M \rightarrow F, \quad x \mapsto \sum_{k=0}^d (T_k - S_k) \circ \text{diag}_k(x).$$

This function is constantly zero on M , hence any of its derivatives is zero also. On the other hand ϕ itself is a polynomial function, and so the order $\ell = d$ derivative formula gives that for any $x \in M$,

$$0 = \phi^{(d)}(x) = [(T_d - S_d) \circ \text{diag}_d]^{(d)}(x) = d! \text{Sym}_d(T_d - S_d),$$

so that $T_d - S_d$ is in the kernel of symmetrization. On the one hand this kernel is precisely the space of all alternating bounded d -linear functions, on the other hand we started with two symmetric bounded d -linear functions, so that their difference is both alternating and symmetric; which implies that this difference is zero¹³. Now arguing as in the single variable case establishes that for any n , $T_n = S_n$, and that d is independent of $T_\bullet$, concluding the solution.

¹³The finite dimensional version of this statement was part of the first problem in this problem set.

Follow-up Exercise 1 Let E, F be normed spaces, $k, \ell \in \{0, 1, 2, 3\}$, $T_k \in \mathcal{BL}_k(E; F)$. Compute $T_k^{(\ell)}$. You may assume that E and F are finite dimensional if you so prefer. **There is nothing to prove in this problem.**

Follow-up Exercise 2 Prove all statements left to you in the solution above.

8 PSet 8, Problem 2

Let $M \subseteq \mathbb{R}^n$ and $N \subseteq \mathbb{R}^m$ be open, $k \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$, $f \in C^k(M; N)$, $x \in M$. Suppose $n \geq m = \text{rank}(f'(x))$.

- (i) There is at least one $g \in C^k(\tilde{M}; \mathbb{R}^{n-m})$ for some open subset $\tilde{M} \subseteq M$ with $x \in \tilde{M}$ such that (f, g) is a local C^k diffeomorphism near x .
- (ii) There is at most one $g \in C^k(\tilde{M}; \mathbb{R}^{n-m})$ for some open subset $\tilde{M} \subseteq M$ with $x \in \tilde{M}$ such that (f, g) is a local C^k diffeomorphism near x .

The first claim is correct, however the second claim is not. Indeed only by focusing on affine functions g (which are in particular C^∞), we'll be able to display a family. Recall that an affine function is a polynomial function of degree at most 1.

Claim: Let $M \subseteq \mathbb{R}^n$ and $N \subseteq \mathbb{R}^m$ be open, $k \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$, $f \in C^k(M; N)$, $x \in M$. Suppose $n \geq m = \text{rank}(f'(x))$. Then there is an affine $g : M \rightarrow \mathbb{R}^{n-m}$ such that

- (i) $g(x) = 0$, and
- (ii) For any sufficiently small affine perturbation $\bar{g}$ of g such that $\bar{g}(x) = 0$, there is an open subset $\tilde{M} \subseteq M$ with $x \in \tilde{M}$ such that $(f, \bar{g})$ is a local C^k diffeomorphism near x .

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Here by $\bar{g}$ being a "sufficiently small affine perturbation" of g we mean that the linear part of $\bar{g}$ is close to the linear part of g relative to the operator norm. Further, we leave the question "how close is close enough" open for now, implying that this question will be answered,

at least implicitly, in the proof. Finally recall that the space of linear transformations (and hence affine functions) between finite dimensional normed spaces itself is finite dimensional, hence the specific choice of the norm does not affect the truth of the claim; although it will certainly affect the answer to the more specific question of how close is close enough. There are contexts where more specific claims and estimates are needed, however this problem asks existence and uniqueness, and so for nonuniqueness certainly a "softer" approach is appropriate.

Proof: Let f and $x \in M$ be as in the statement. Then $f'(x)$ can be considered to be an $m \times n$ matrix. We are given that this matrix has rank exactly m , and that $m \leq n$. Thus by **row reduction**, there are two square matrices L_x, P_x such that

- (i) L_x is an $m \times m$ lower triangular invertible matrix,
- (ii) P_x is an $n \times n$ permutation matrix (that is, P_x has exactly one 1 in each row and column, and all other entries are 0),
- (iii) $f'(x) = L_x \begin{pmatrix} I_m & 0_{m,n-m} \end{pmatrix} P_x$, where I_m is the $m \times m$ identity matrix and $0_{m,n-m}$ is the $m \times (n-m)$ zero matrix.

In more abstract language, this matrix factorization corresponds to the fact that in some appropriate bases β and γ , the matrix representation of $f'(x)$ is the row and column reduced echelon matrix involved:

$$\mathcal{M}_{\gamma \leftarrow \beta}(f'(x)) = \begin{pmatrix} I_m & 0_{m,n-m} \end{pmatrix}.$$

The point of the construction to follow is that, again in appropriate bases, g is so that $g'(x)$ supplies the remaining $n-m$ pivots, ensuring invertibility of $(f, g)'(x)$.

For any $A \in \mathcal{L}(\mathbb{R}^{n-m}; \mathbb{R}^{n-m})$, define $g_A : M \rightarrow \mathbb{R}^{n-m}$ by

$$g_A : \bar{x} \mapsto A \begin{pmatrix} 0_{n-m,m} & I_{n-m} \end{pmatrix} P_x (\bar{x} - x),$$

where $0_{n-m,m}$ is the $(n - m) \times m$ zero matrix and I_{n-m} is the $(n - m) \times (n - m)$ identity matrix. Then note that

$$g'_A(x) = A \begin{pmatrix} 0_{n-m,m} & I_{n-m} \end{pmatrix} P_x,$$

and consequently

$$\begin{aligned} (f, g_A)'(x) &= \begin{pmatrix} L_x & 0_{m,n-m} \\ 0_{n-m,m} & A \end{pmatrix} \begin{pmatrix} I_m & 0_{m,n-m} \\ 0_{n-m,m} & I_{n-m} \end{pmatrix} P_x \\ &= \begin{pmatrix} L_x & 0_{m,n-m} \\ 0_{n-m,m} & A \end{pmatrix} I_n P_x \\ &= \begin{pmatrix} L_x & 0_{m,n-m} \\ 0_{n-m,m} & A \end{pmatrix} P_x, \end{aligned}$$

whence

$$\det((f, g_A)'(x)) = \pm \det(L_x) \det(A),$$

and so $(f, g_A)'(x)$ is invertible iff A is invertible. Note that there are many invertible A 's, for instance based on a problem in [problem set 5](#), for any H with $|H|_{\text{op}} < 1$, $A = I + H$ is invertible; more concretely one can take A to be the diagonal matrix with the $(1, 1)$ entry $1 + 1/n$ and all other entries 1, for any $n \in \mathbb{Z}_{\geq 2}$. For any such A , we can then apply the Inverse Function Theorem to (f, g_A) to obtain a local inverse ϕ_A that is uniquely determined, but this local inverse in principle will depend on A .

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Follow-up Exercise Suppose f is a function and $\bar{f}$ is a sufficiently small perturbation of f . Formalize and prove statements that guarantee that if the Inverse or Implicit Function Problem associated to f has a solution φ , then the Inverse or Implicit Function Problem associated to $\bar{f}$ has a solution $\bar{\varphi}$, and $\bar{\varphi}$ itself is a sufficiently small perturbation of φ .

9 PSet 9, Problem 4

Let V be a finite dimensional inner product space, $S \subseteq V$ be a Jordan contentable subset with positive Jordan content, W be a finite dimensional normed space, $f \in R(S; W)$. Suppose S is compact and path-connected, and f is continuous.

- (i) There is an $x_0 \in S$ such that $f(x_0) = \mathbb{M}(S, f)$.
- (ii) There is an $x_0 \in S$ such that $f(x_0) = \mathbb{M}(S, f)$, if $\dim(W) = 1$.

This problem is about the "integral form" of the mean value theorem. Based on this observation, our expectation ought to be that the statement holds for scalar valued functions but not necessarily for vector valued functions in general. Indeed this expectation is accurate.

First we state an auxiliary lemma that will allow us to reduce the statement to the case of functions between spaces of tuples:

Lemma 4: Let V be a finite dimensional inner product space, $S \subseteq V$ be a Jordan contentable subset, W, X be finite dimensional normed spaces. Then for any $f \in R(S; W)$ and any $A \in \mathcal{L}(W; X)$,

- (i) $A \circ f \in R(S; X)$,
- (ii) $A \left(\int_S f(x) dx \right) = \int_S A \circ f(x) dx$.
- (iii) If S has positive Jordan content, then $\mathbb{M}(S, A \circ f) = A(\mathbb{M}(S, f))$.

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The first part of the lemma is a special case of an exercise mentioned in class. The second part follows from the equality at the level of Darboux sums by the linearity and continuity of A , and the last part follows from the second part.

Claim: Let V be a finite dimensional inner product space, $S \subseteq V$ be a Jordan contentable subset with positive Jordan content, W be a finite dimensional normed space, $f \in R(S; W)$. Suppose S is compact and path-connected, and f is continuous. If $\dim(W) = 1$, then there is an $x_0 \in S$ such that $f(x_0) = \mathbb{M}(S, f)$, if $\dim(W) = 1$.

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Proof: By the lemma, we may assume that $V = \mathbb{R}^n$ and $W = \mathbb{R}$. As S is compact and f is continuous, by the Maximum Principle, for any $x \in S$,

$$\min(f) \leq f(x) \leq \max(f).$$

By the monotonicity of Riemann integrals,

$$\text{jor}(S) \min(f) \leq \int_S f(x) d(x) \leq \text{jor}(S) \max(f).$$

Thus $\mathbb{M}(S, f) \in [\min(f), \max(f)]$. As S is path connected and f is continuous, this interval is the image of S under f , hence there must be a point $x_0 \in S$ such that $f(x_0) = \mathbb{M}(S, f)$.

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For vector valued functions the claim is not correct, and we'll need to construct a counterexample. Consider a general continuous function $f = (f_1, \dots, f_m) : S \rightarrow \mathbb{R}^m$ defined on the standard aligned box

$$S = \underbrace{[0, 1] \times [0, 1] \times \cdots \times [0, 1]}_{n \text{ many}} \subseteq \mathbb{R}^n.$$

We have already established above that for each component function f_i separately the statement holds. Also note that by the definition of the integral,

$$\mathbb{M}(S; f) = (\mathbb{M}(S; f_1), \dots, \mathbb{M}(S; f_m)).$$

Thus the general strategy will be to engineer the components f_i so that the level sets of means of components are disjoint, so that averages are achieved, but not simultaneously, at any point in S . To that end we'll consider a family of "pyramid" functions. (It's best to follow the construction of pyramid functions using sketches and assuming $n = 2$ or $n = 3$.)

For any $\xi = (\xi_1, \xi_2, \dots, \xi_n) \in S \cap \mathbb{R}_{>0}^n$, put

$$B_\xi = [0, \xi_1] \times \cdots \times [0, \xi_n] \subseteq S,$$

and define

$$\wp_\xi : S \rightarrow \mathbb{R}_{\geq 0},$$

$$(x_1, x_2, \dots, x_n) \mapsto \begin{cases} 1 - \frac{x_i}{\xi_i}, & \text{if } (x_1, \dots, x_n) \in B_\xi, \\ & \text{and } x_i \geq \max_{j \neq i} x_j \\ 0, & \text{if } (x_1, \dots, x_n) \notin B_\xi \end{cases}.$$

$\wp_\xi$ then is the function that is zero outside the subbox B_ξ and on B_ξ it is so that the subset

$$\{(x, h) \in B_\xi \times \mathbb{R}_{\geq 0} \mid h \leq \wp_\xi(x)\}$$

is the **pyramid** whose apex is $(0, 0, \dots, 0, 1)$ and whose base is the subbox B_ξ .

The following lemma summarizes the key properties of $\wp_\xi$ we'll need. Denote by T_ξ the collection of all $\eta \in \mathbb{R}^n$ such that for any $\tau_\eta^{-1}(B_\xi) \subseteq S$, where τ_η is the translation $x \mapsto x + \eta$. For any $\eta \in T_\xi$, put $\wp_{\xi, \eta} = \wp_\xi \circ \tau_\eta : S \rightarrow \mathbb{R}_{\geq 0}$, $B_{\xi, \eta} = \tau_\eta^{-1}(B_\xi)$. Then $\wp_{\xi, \eta}$ too is a pyramid function that is zero outside the standard aligned subbox $B_{\xi, \eta}$.

Lemma 5: For any $\xi = (\xi_1, \xi_2, \dots, \xi_n) \in S \cap \mathbb{R}_{>0}^n$, and for any $\eta \in T_\xi$,

- (i) $\wp_{\xi,\eta} : S \rightarrow \mathbb{R}_{\geq 0}$ is continuous (and hence Riemann integrable).
- (ii) $\wp_{\xi,\eta}^{-1}(\mathbb{M}(S; \wp_{\xi,\eta})) \subseteq B_{\xi,\eta}$.

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Proof (of lemma): First let us verify the continuity of $\wp_{\xi}$. For any $i \in \{1, 2, \dots, n\}$, denote by $B_{\xi,i}$ the subset of B_{ξ} defined by

$$B_{\xi,i} = \{(x_1, \dots, x_n) \in B_{\xi} \mid x_i \geq \max_{j \neq i} x_j\}.$$

Then these subsets are pairwise disjoint and $\bigcup_{i=1}^n \text{clos}(B_{\xi,i}) = B_{\xi}$. On each $B_{\xi,i}$, $\wp_{\xi}$ is an affine, hence continuous function, that depends only on the largest component of the input vector. Two $B_{\xi,i}$'s intersect exactly at inputs where two largest components of the input vectors are equal, so that the two affine functions coincide, so that $\wp_{\xi}$ is continuous on B_{ξ} . On the boundary of B_{ξ} that is in the interior of S , $\wp_{\xi}$ vanishes, so $\wp_{\xi}$ is continuous on S .

Next up we'll establish that the fact that the mean level set of $\wp_{\xi,\eta}$ is a (nonempty) subset of $B_{\xi,\eta}$. First, $\wp_{\xi,\eta}$ is a nonnegative function, and since it is nonzero on a set of positive Jordan content (namely $B_{\xi,\eta}$), its mean is a positive number. Since outside $B_{\xi,\eta}$, the function is zero, $\wp_{\xi,\eta}$ can assume its mean only inside $B_{\xi,\eta}$.

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In the context of the previous lemma, we could have explicitly computed the mean, as well as the mean level set of $\wp_{\xi,\eta}$; for the rest of the argument we'll not need these more specific pieces of information.

Claim: Let V be a finite dimensional inner product space, $S \subseteq V$ be a Jordan contentable subset with positive Jordan content, W be a finite dimensional normed space, $f \in R(S; W)$. Suppose S is compact and path-connected, and f is continuous. If $\dim(W) \geq 2$, then depending

on the function f , the level set $\{x \in S \mid f(x) = \mathbb{M}(S, f)\}$ may be empty or nonempty.

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Proof: First note that if f is a constant function, then for any $x \in S$, $f(x)$ is equal to the mean of f . Thus it suffices to construct counterexamples where the mean is not achieved by the function. Let

$$\xi = \frac{1}{2m}(1, 1, \dots, 1),$$

and for any $j \in \{1, 2, \dots, m\}$, let

$$\eta_j = \frac{j-1}{m}(1, 0, \dots, 0).$$

Finally consider the function $f = (\varphi_{\xi, \eta_1}, \varphi_{\xi, \eta_2}, \dots, \varphi_{\xi, \eta_m}) : S \rightarrow \mathbb{R}_{\geq 0}$. Then no two components can achieve their own respective means at the same point in S .

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Follow-up Exercise Let V be a finite dimensional inner product space, $S \subseteq V$ be a Jordan contentable subset with positive Jordan content, W be a finite dimensional normed space, $f \in R(S; W)$.

Consider the following additional hypotheses:

- (A) S is compact.
- (B) S is convex.
- (C) S is path connected.
- (D) f is continuous.
- (E) f takes finitely many values.
- (F) f is constant.
- (G) W is one-dimensional.

Perform the following tasks:

- (i) Classify all collections of hypotheses that guarantee there is an $x_0 \in S$ such that $f(x_0) = \mathbb{M}(S, f)$.
- (ii) For any $n \in \mathbb{Z}_{\geq 1}$, classify all collections of hypotheses that guarantee that the set of all those $x_0 \in S$ such that $f(x_0) = \mathbb{M}(S, f)$ is Jordan measurable and has positive n -dimensional Jordan content.

Follow-up Exercise 2 There is nothing to prove in this problem.

Let $\xi = (\xi_1, \xi_2, \dots, \xi_n) \in S \cap \mathbb{R}_{>0}^n$, and $\eta \in T_\xi$. Then

- (i) $\mathbb{M}(S; \varphi_{\xi, \eta}) = \int_S \varphi_{\xi, \eta}(x) dx = \frac{\text{vol}^n(B_\xi)}{n+1}$.
- (ii) Compute the standard aligned subbox $M_{\xi, \eta}$ of $B_{\xi, \eta}$ such that the level set $\varphi_{\xi, \eta}^{-1}(\mathbb{M}(S; \varphi_{\xi, \eta})) \subseteq B_{\xi, \eta}$ is the boundary of $M_{\xi, \eta}$.

10 PSet 10, Problem 2

There is nothing to prove in this problem. In this problem we consider differential 1-forms on $\mathbb{R}^2$. For any $v = (v_1, \dots, v_6) \in \mathbb{R}^6$, define

$$\omega_v = (v_1 + v_2x + v_3y)dx + (v_4 + v_5x + v_6y)dy.$$

- (i) Compute the set of all $v \in \mathbb{R}^6$ such that ω_v is exact.
- (ii) Compute the set of all $v \in \mathbb{R}^6$ such that ω_v is closed.

Condition for Exactness For the first part, we want a C^1 differential 0-form f such that

$$df = \omega_v.$$

Writing this out, this means we need

$$\frac{\partial f}{\partial x} = v_1 + v_2x + v_3y, \quad \frac{\partial f}{\partial y} = v_4 + v_5x + v_6y.$$

In this problem the set on which we are considering is the whole of $\mathbb{R}^2$, so we may safely use the standard Calc III methods. Taking the antiderivative of the first equation with respect to x , we have that for some $\phi \in C^1$ that is independent of x ,

$$f = v_1x + \frac{v_2}{2}x^2 + v_3xy + \phi(y).$$

Taking then the y -partial of this expression, we obtain

$$v_3x + \frac{d\phi}{dy}(y) = v_4 + v_5x + v_6y.$$

Rearranging,

$$\frac{d\phi}{dy}(y) - v_6 y = v_4 + (v_5 - v_3)x.$$

The LHS is independent of x and the RHS is independent of y , hence neither is dependent on either, that is, $v_5 = v_3$ and

$$\frac{d\phi}{dy}(y) = v_4 + v_6 y,$$

so that for some constant $c \in \mathbb{R}$,

$$\phi(y) = v_4 y + \frac{v_6}{2} y^2 + c.$$

Finally putting these together, assuming $v_3 = v_5$, we obtain a C^1 differential 0-form f such that $df = \omega_v$:

$$f = v_1 x + \frac{v_2}{2} x^2 + v_3 xy + v_4 y + \frac{v_6}{2} y^2;$$

indeed any other solution $dg = \omega_v$ is a translation of f by a constant. Consequently ω_v is exact iff $v_3 = v_5$.

Condition for Closedness For closedness we simply need to take the exterior derivative and arrange the coefficient vector v so that $d\omega_v = 0$.

$$\begin{aligned} d\omega_v &= d(v_1 + v_2 x + v_3 y) \wedge dx + d(v_4 + v_5 x + v_6 y) \wedge dy \\ &= v_3 dy \wedge dx + v_5 dx \wedge dy \\ &= (v_5 - v_3) dx \wedge dy. \end{aligned}$$

Hence ω_v is closed iff $v_3 = v_5$.

Relation between Closedness and Exactness In a different problem in this problem set you are asked to verify that exactness implies closedness, on any open subset of tuples. As the calculation above shows, often coming up with conditions for exactness is a lot more complicated than coming up with conditions of closedness. As a first step, it thus is rather useful to first establish conditions for closedness, thus simplifying the space for search of exactness right from the get-go.

Follow-up Exercise Compute the collection of all open subsets of M such that for any $v \in \mathbb{R}^6$, any

$$\omega_v = (v_1 + v_2x + v_3y)dx + (v_4 + v_5x + v_6y)dy$$

with $v_3 = v_5$. considered as a differential 1-form on M , is exact.

Follow-up Exercise 2 Let $M = \mathbb{R}^n$. A **polynomial differential form** on M is a differential form each of whose coefficient functions is a polynomial function. Compute all exact and closed polynomial differential forms on $\mathbb{R}^n$.

11 PSet 11, Problem 3

There is nothing to prove in this problem. Consider the following $\omega \in C^\infty\text{-Form}^1(\mathbb{R}^3)$, $\gamma \in C^\infty(\mathbb{R}; \mathbb{R}^3)$ and $\alpha, \beta \in \mathbb{R}$:

I: $\omega = xdx + ydy$, $\gamma : t \mapsto (t^2, t^3, 0)$, $\alpha = 0$, $\beta = 1$.

II: $\omega = xdx + ydy$, $\gamma : t \mapsto ((1-t)^2, (1-t)^3, 0)$, $\alpha = 0$, $\beta = 1$.

III: $\omega = xzdx - x^2dz$, $\gamma : t \mapsto (\cos t, 0, \sin t)$, $\alpha = 0$, $\beta = \pi/2$.

IV: $\omega = xzdx - x^2dz$, $\gamma : t \mapsto (\cos(\pi t), 0, \sin(\pi t))$, $\alpha = 0$, $\beta = 1/2$.

V: $\omega = x^2dx + y^2dy + dz$, $\gamma : t \mapsto (\cos(2\pi t), \sin(2\pi t), t-t^2)$, $\alpha = 1$, $\beta = 0$.

VI: $\omega = \sin zdx + \cos zdy + y^2dz$, $\gamma : t \mapsto (\cos t, \sin t, t)$, $\alpha = 2\pi$, $\beta = 0$

For each $(\omega, \gamma, \alpha, \beta)$, perform the following tasks:

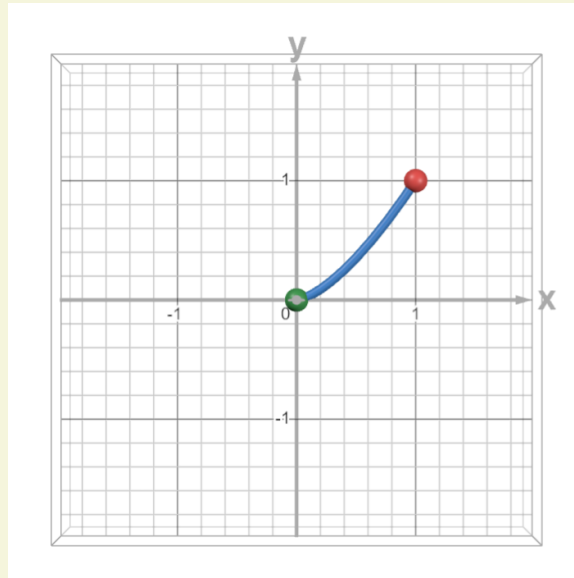
- (i) For $I_{\alpha, \beta}$ the compact interval with endpoints α and β , sketch $\gamma(I_{\alpha, \beta})$.
- (ii) Compute $f_{\omega, \gamma} \in C^\infty(\mathbb{R}; \mathbb{R})$ such that $\gamma^*(\omega) = f_{\omega, \gamma} dt$.
- (iii) Compute $\int_{\alpha}^{\beta} f_{\omega, \gamma}(t) dt$.

We first note that any triple (γ, α, β) as given above defines a unique 1-cell; namely $[\gamma; \beta \leftarrow \alpha]$. We are asked to sketch a visual representative for each of these 1-cells and then compute the integrals $\langle [\gamma; \beta \leftarrow \alpha] | \omega \rangle$. For sketches we'll use Desmos 3D; here is the link to the interactive graph:

<https://www.desmos.com/3d/xpl6iol1lo>.

I $\omega = xdx + ydy$, $\gamma : t \mapsto (t^2, t^3, 0)$, $\alpha = 0$, $\beta = 1$.

The sketch of $\gamma(l_{\alpha,\beta})$ looks like this:



Here, and in any other sketch that follows, the orientation is from the green endpoint to the red endpoint.

We next compute the integrand using the synthetic approach to pullback:

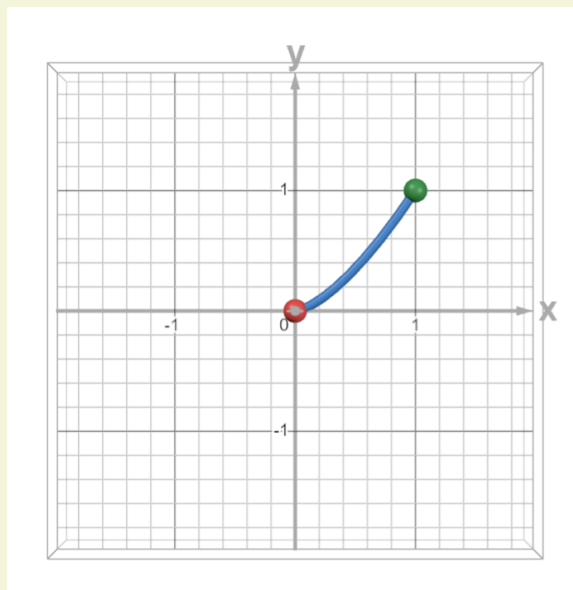
$$\begin{aligned}
 \gamma^*(\omega) &= \gamma^*(xdx + ydy) \\
 &= t^2d(t^2) + t^3d(t^3) \\
 &= 2t^3dt + 3t^5dt \\
 &= \underbrace{(2t^3 + 3t^5)}_{=f_{\omega,\gamma}(t)} dt.
 \end{aligned}$$

Finally the integral is computed in a straightforward fashion as follows:

$$\begin{aligned}
 \langle [\gamma; \beta \leftarrow \alpha] | \omega \rangle &= \int_{\alpha}^{\beta} \gamma^*(\omega) \\
 &= \int_{\alpha}^{\beta} f_{\omega, \gamma}(t) dt \\
 &= \int_0^1 (2t^3 + 3t^5) dt \\
 &= \left[t^4/2 + t^6/2 \right] \Big|_{t=0}^1 \\
 &= 1
 \end{aligned}$$

We'll handle the other parts similarly with minimal commentary, as the procedure is fairly routine.

II $\omega = xdx + ydy$, $\gamma : t \mapsto ((1-t)^2, (1-t)^3, 0)$, $\alpha = 0$, $\beta = 1$.



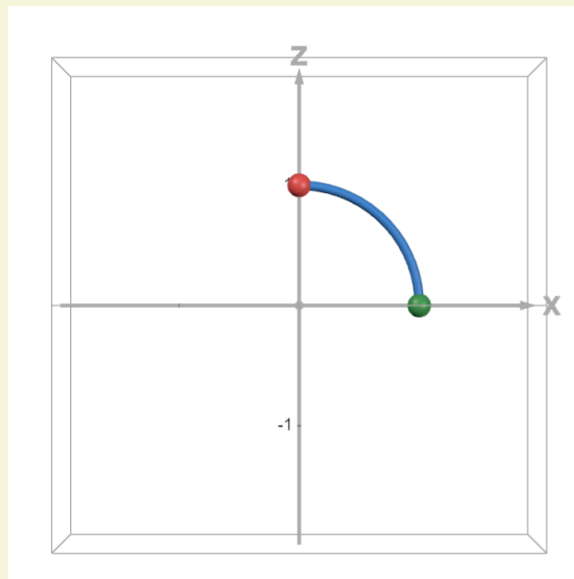
We note that this 1-cell is *not the same* as the 1-cell in the previous part; indeed the orientations are opposite.

$$\begin{aligned}
 \gamma^*(\omega) &= \gamma^*(x dx + y dy) \\
 &= (1-t)^2 d((1-t)^2) + (1-t)^3 d((1-t)^3) \\
 &= -2(1-t)^3 dt - 3(1-t)^5 dt \\
 &= \underbrace{-(2(1-t)^3 + 3(1-t)^5)}_{=f_{\omega,\gamma}(t)} dt.
 \end{aligned}$$

Instead of computing the integral directly, by the u-substitution $u = 1 - t$ one can see that the end result is -1 times the value of the previous integral, hence

$$\langle [\gamma; \beta \leftarrow \alpha] | \omega \rangle = -1.$$

III $\omega = xz dx - x^2 dz$, $\gamma : t \mapsto (\cos t, 0, \sin t)$, $\alpha = 0$, $\beta = \pi/2$.



$$\begin{aligned}
\gamma^*(\omega) &= \gamma^*(xzdx - x^2dz) \\
&= (\cos t)(\sin t)d(\cos t) - (\cos t)^2d(\sin t) \\
&= -(\cos t)(\sin t)^2dt - (\cos t)^3dt \\
&= -\cos t((\sin t)^2 + (\cos t)^2)dt \\
&= \underbrace{-\cos t}_{=f_{\omega,\gamma}(t)}dt.
\end{aligned}$$

$$\begin{aligned}
\langle [\gamma; \beta \leftarrow \alpha] | \omega \rangle &= \int_{\alpha}^{\beta} \gamma^*(\omega) \\
&= \int_{\alpha}^{\beta} f_{\omega,\gamma}(t)dt \\
&= \int_0^{\pi/2} (-\cos t)dt \\
&= [-\sin t]_{t=0}^{\pi/2} \\
&= -1
\end{aligned}$$

IV $\omega = xzdx - x^2dz$, $\gamma : t \mapsto (\cos(\pi t), 0, \sin(\pi t))$, $\alpha = 0$, $\beta = 1/2$.

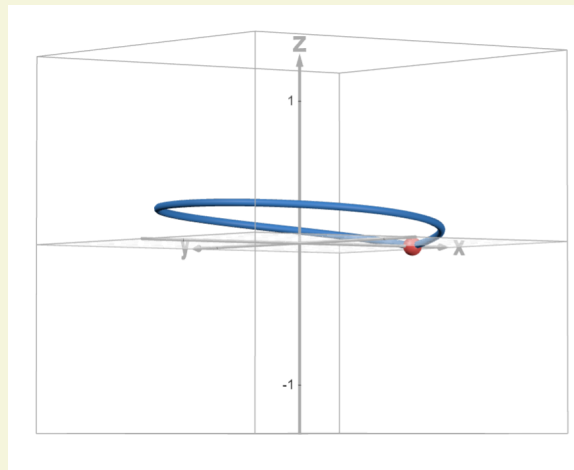
This 1-cell *exactly the same* as the 1-cell in the previous part. Note that while the parameterizations are different, two 1-cells are the same insofar as they are the same as 1-currents (aka real valued linear functions on differential 1-forms).

Note that this *does not mean* that the associated pullbacks will be the same:

$$\begin{aligned}
 \gamma^*(\omega) &= \gamma^*(xzdx - x^2dz) \\
 &= (\cos(\pi t))(\sin(\pi t))d(\cos(\pi t)) - (\cos(\pi t))^2 d(\sin(\pi t)) \\
 &= -\pi(\cos(\pi t))(\sin(\pi t))^2 dt - \pi(\cos(\pi t))^3 dt \\
 &= -\pi \cos(\pi t) ((\sin(\pi t))^2 + (\cos(\pi t))^2) dt \\
 &= \underbrace{-\pi \cos(\pi t)}_{=f_{\omega, \gamma}(t)} dt.
 \end{aligned}$$

Again by a u-substitution we obtain -1 as the value of the associated integral.

V $\omega = x^2 dx + y^2 dy + dz$, $\gamma : t \mapsto (\cos(2\pi t), \sin(2\pi t), t - t^2)$, $\alpha = 1$, $\beta = 0$.



We first compute the pullback directly:

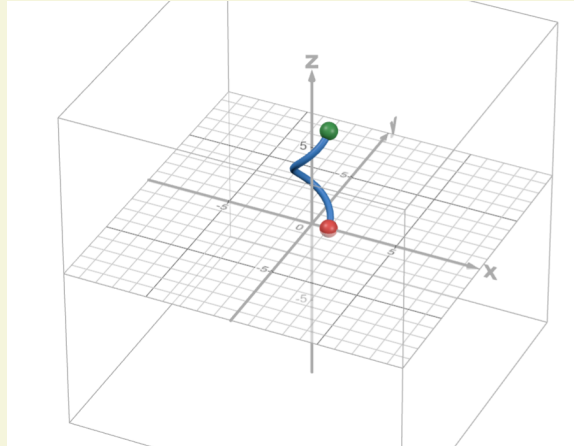
$$\begin{aligned}
\gamma^*(\omega) &= \gamma^*(x^2 dx + y^2 dy + dz) \\
&= (\cos(2\pi t))^2 d(\cos(2\pi t)) + (\sin(2\pi t))^2 d(\sin(2\pi t)) + d(t - t^2) \\
&= -2\pi(\cos(2\pi t))^2(\sin(2\pi t))dt + 2\pi(\sin(2\pi t))^2(\cos(2\pi t))dt + (1 - 2t)dt \\
&= [2\pi(\sin(2\pi t))^2 \cos(2\pi t) - 2\pi(\cos(2\pi t))^2 \sin(2\pi t) + 1 - 2t] dt \\
&= [2\pi \sin(2\pi t) \cos(2\pi t)(\sin(2\pi t) - \cos(2\pi t)) + 1 - 2t] dt \\
&= \underbrace{[\pi \sin(4\pi t)(\sin(2\pi t) - \cos(2\pi t)) + 1 - 2t]}_{=f_{\omega,\gamma}(t)} dt.
\end{aligned}$$

Here one can straightforwardly integrate the function $f_{\omega,\gamma}$ (using integration by parts, among other things). However, instead of this we'll use the single variable Fundamental Theorem of Calculus, together with the fact that pullbacks commute with exterior derivatives. First note that $\omega = x^2 dx + y^2 dy + dz$ is exact; indeed for the C^∞ differential 0-form $\phi = x^3/3 + y^3/3 + z$, we have $d\phi = \omega$. Then

$$\gamma^*(\omega) = \gamma^*(d\phi) = d\gamma^*(\phi) = d(\phi \circ \gamma) = (\phi \circ \gamma)'(t)dt;$$

so that $(\phi \circ \gamma)' = f_{\omega,\gamma}$. Hence since $\phi \circ \gamma(1) = \phi \circ \gamma(0)$, the associated integral is 0.

VI $\omega = \sin z dx + \cos z dy + y^2 dz$, $\gamma : t \mapsto (\cos t, \sin t, t)$, $\alpha = 2\pi$, $\beta = 0$



$$\begin{aligned}
 \gamma^*(\omega) &= \gamma^*(\sin z dx + \cos z dy + y^2 dz) \\
 &= \sin t d(\cos t) + \cos t d(\sin t) + (\sin t)^2 dt \\
 &= -(\sin t)^2 dt + (\cos t)^2 dt + (\sin t)^2 dt \\
 &= \underbrace{(\cos t)^2}_{=f_{\omega,\gamma}(t)} dt.
 \end{aligned}$$

To compute the associated integral, in this case there are no shortcuts and hence one would need to use integration by parts by setting $u = \cos t$, $dv = \cos t dt$, and then applying integration by parts again to obtain $-\pi$ as the final answer.

12 PSet 12, Problem 5

Let $M \subseteq \mathbb{R}^2$ be an open subset not containing $(0, 0)$, and consider the following C^∞ differential 1-form on M :

$$\omega = \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy.$$

- (i) ω is closed.
- (ii) ω is exact.

We'll argue that ω is closed on any open subset $M \subseteq \mathbb{R}^2$ not containing the origin (this is the only point where ω is not defined). On the other hand, depending on how the origin and M is situated ω may fail to be exact.

Claim: Let $M \subseteq \mathbb{R}^2$ be an open subset not containing $(0, 0)$ and consider the C^∞ differential 1-form $\omega = \frac{-ydx+xdy}{x^2+y^2}$ on M . Then ω is closed.

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Proof: To establish closedness we need to verify that the exterior derivative of ω vanishes. A straightforward calculation, say using the synthetic approach, guarantees exactly this:

$$\begin{aligned}
d\omega &= d\left(\frac{-ydx + xdy}{x^2 + y^2}\right) \\
&= d\left(\frac{-y}{x^2 + y^2}dx + \frac{x}{x^2 + y^2}dy\right) \\
&= d\left(\frac{-y}{x^2 + y^2}\right) \wedge dx + d\left(\frac{x}{x^2 + y^2}\right) \wedge dy \\
&= \left[-\frac{\partial}{\partial y}\left(\frac{-y}{x^2 + y^2}\right) + \frac{\partial}{\partial x}\left(\frac{x}{x^2 + y^2}\right)\right] dx \wedge dy \\
&= \left[-\frac{-(x^2 + y^2) + y \cdot 2y}{(x^2 + y^2)^2} + \frac{(x^2 + y^2) + x \cdot 2x}{(x^2 + y^2)^2}\right] dx \wedge dy \\
&= 0.
\end{aligned}$$

┘

As opposed to closedness, exactness of a form really depends on the *global properties* of the open subset (or more generally manifold) M . To formulate such global properties efficiently, we really need notions from topology we haven't discussed yet in this class; still we can come up with certain conditions on M that would guarantee the exactness or nonexactness of ω on M . The heuristic, both for the claims below as well as why more sophisticated notions are needed, is this: The only singularity of ω is the origin, and insofar as M is so that one complete loop around the origin, while still staying completely within M , is possible, ω will fail to be exact. If there is no complete loop around the origin that is completely within M , then ω will be exact.

We start with a negative claim:

Claim: Let $M \subseteq \mathbb{R}^2$ be an open subset not containing $(0, 0)$ and consider the C^∞ differential 1-form $\omega = \frac{-ydx + xdy}{x^2 + y^2}$ on M . If M contains $\mathbb{R}^2[(0, 0) | \leq \varepsilon] \setminus \{(0, 0)\}$ for some $\varepsilon \in \mathbb{R}_{>0}$, then ω is not exact.

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Proof: Suppose ω is exact. Then if C is a C^1 1-cycle (aka a 1-chain whose boundary is the zero 0-chain), $\langle C | \omega \rangle = 0$ as a consequence of the Fundamental Theorem of Line Integrals. We'll display a certain 1-cycle C such that the associated integral fails to be zero, hence arriving at a contradiction.

Consider $\gamma_\varepsilon : \mathbb{R} \rightarrow M$, $t \mapsto \varepsilon(\cos(t), \sin(t))$. Then a calculation shows that $\gamma_\varepsilon^*(\omega) = dt$, and so

$$\langle [\gamma_\varepsilon; 2\pi \leftarrow 0] | \omega \rangle = 2\pi,$$

a contradiction. ┘

Finally we discuss a claim in the positive:

Claim: Let $M \subseteq \mathbb{R}^2$ be an open subset not containing $(0, 0)$ and consider the C^∞ differential 1-form $\omega = \frac{-ydx + xdy}{x^2 + y^2}$ on M . If M is the image of some open subset of $\mathbb{R}_{>0} \times]0, 2\pi[$ under the polar coordinates map, then ω is exact. ┘

Proof: Recall that the polar coordinates map is defined by $\Phi : (r, \theta) \mapsto (r \cos \theta, r \sin \theta)$. In midterm 2 we had established that Φ is a local C^∞ diffeomorphism away from the line $r = 0$. Since on $\mathbb{R}_{>0} \times]0, 2\pi[$, Φ has a functional inverse, this inverse too is C^∞ . A computation shows that $\Phi^*(\omega) = d\theta$, hence

$$\omega = (\Phi^{-1})^* d\theta = d(\Phi^{-1})^*(\theta).$$

Here by $(\Phi^{-1})^*(\theta)$ we mean the function $(r, \theta) \mapsto \theta$ precomposed with $\Phi^{-1} : (x, y) \mapsto (r, \theta)$. ┘